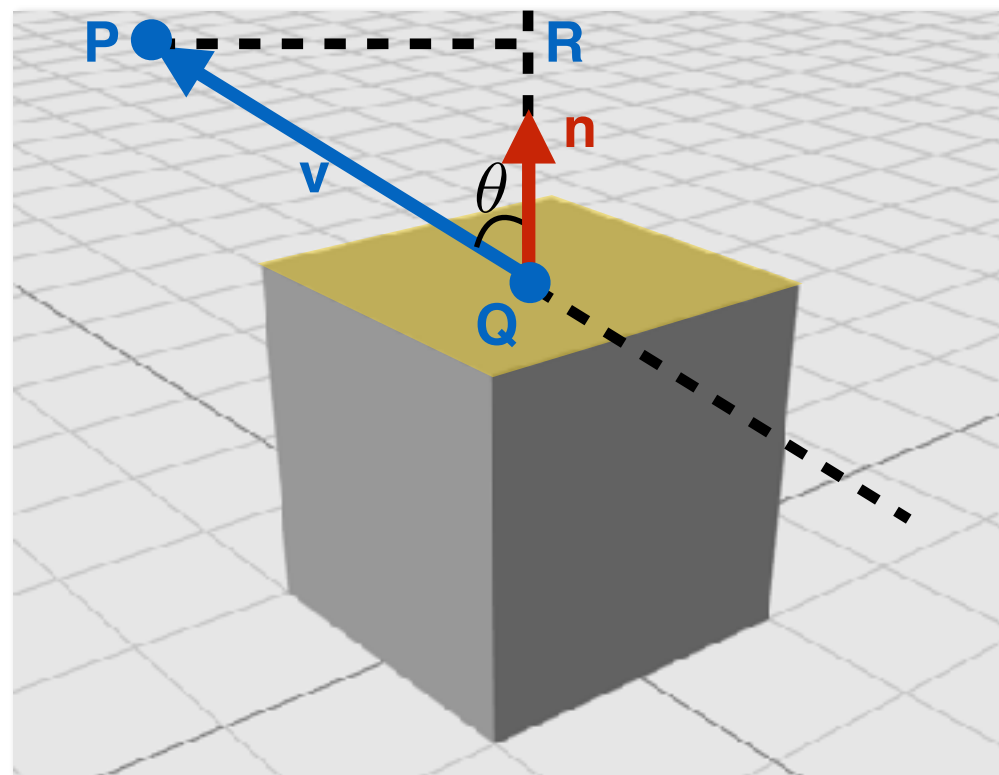


Mathematics in Computer Graphics



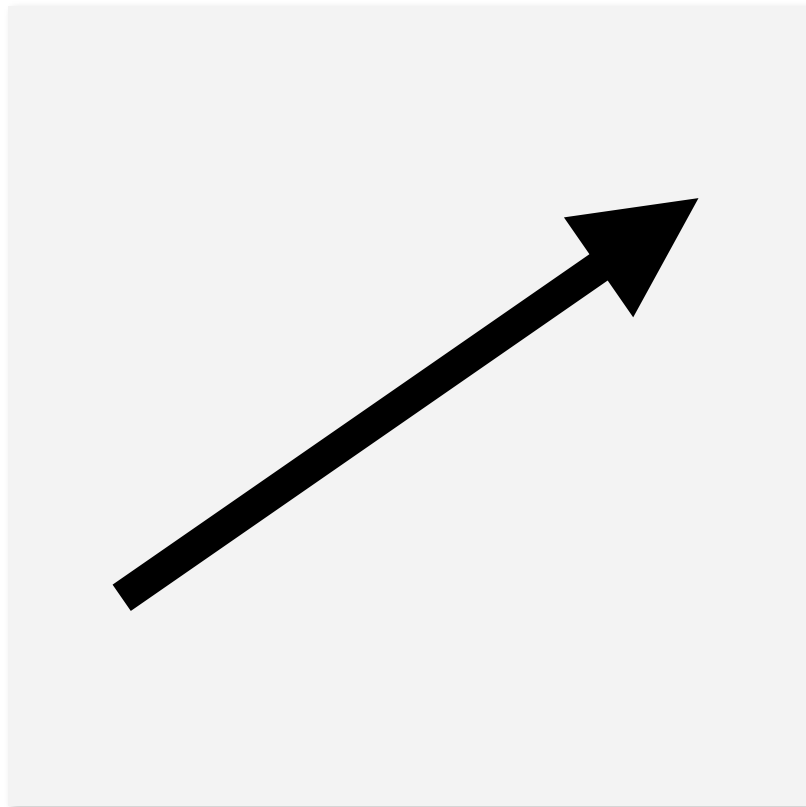
Vectors

- **Vector** - an abstract geometric entity that represents motion between two points (e.g. 10m south)
- **Coordinate vector** - a set of real numbers used to specify a vector (based on a coordinate system)

Vectors

Geometry:

arrows in space



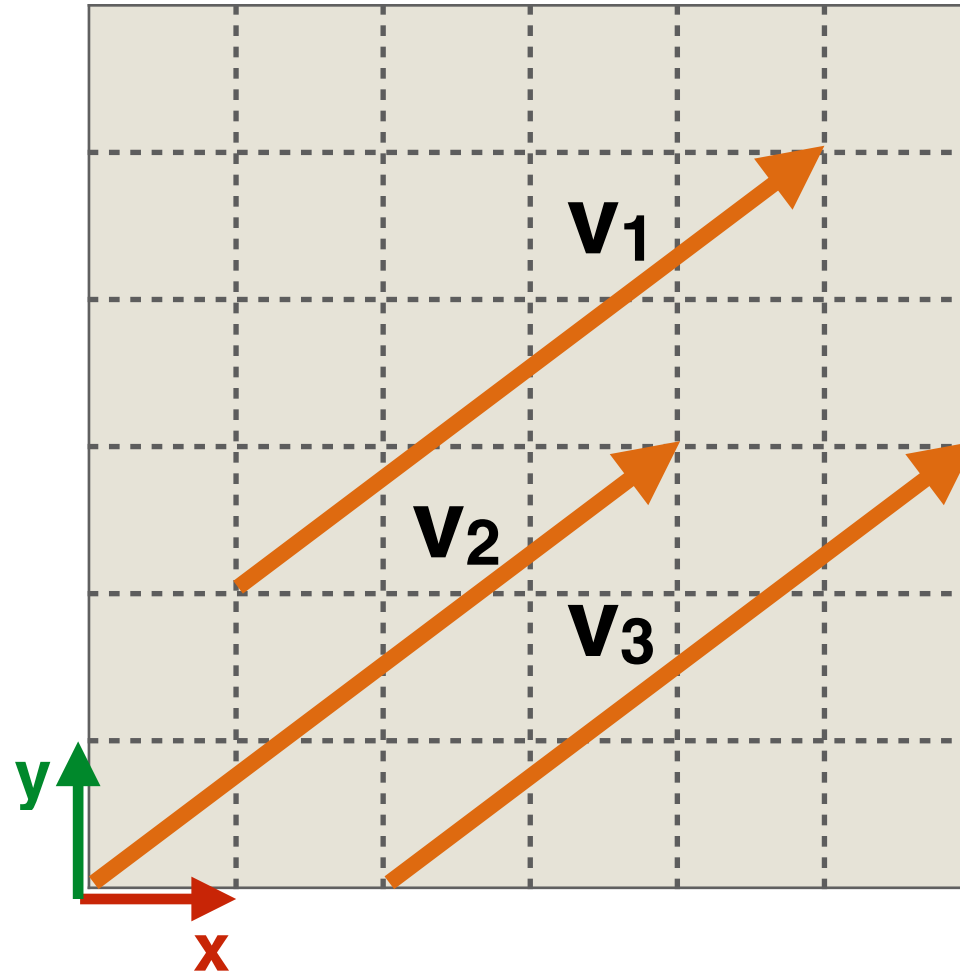
Algebra:

arrays of numbers

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

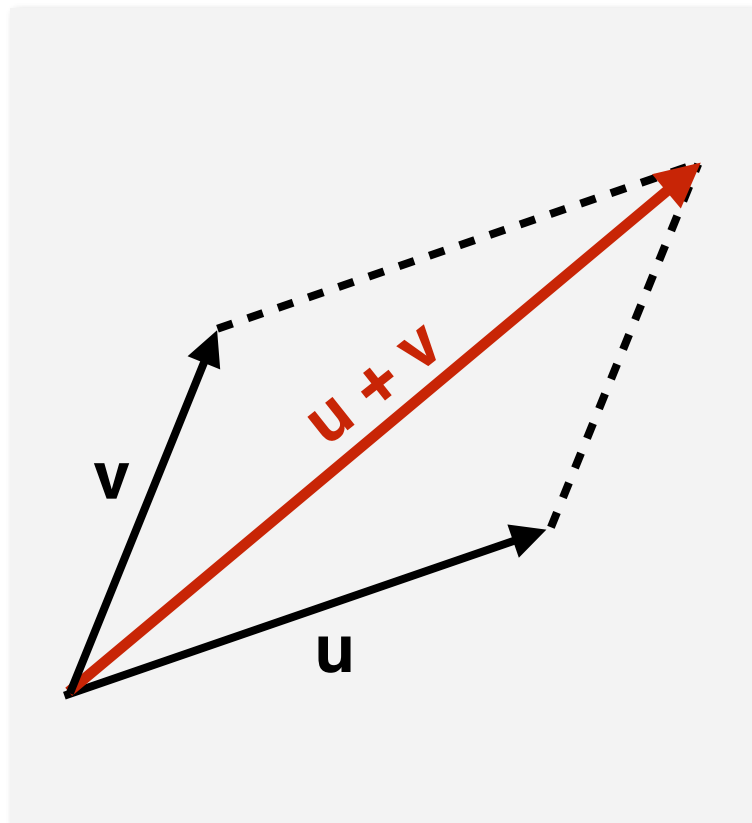
Vectors

- A vector is described by **length** and **direction**
- Represents **displacement**
- Two vectors are **equal** if they have the same length and direction



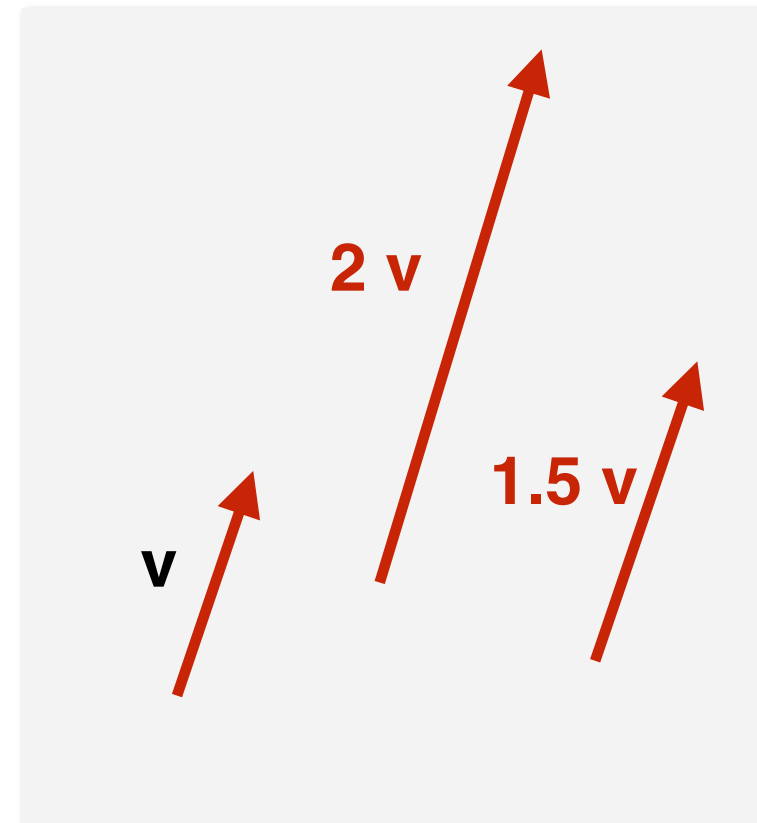
$$\mathbf{v}_1 = \mathbf{v}_2 = \mathbf{v}_3$$

Vector operations



Addition

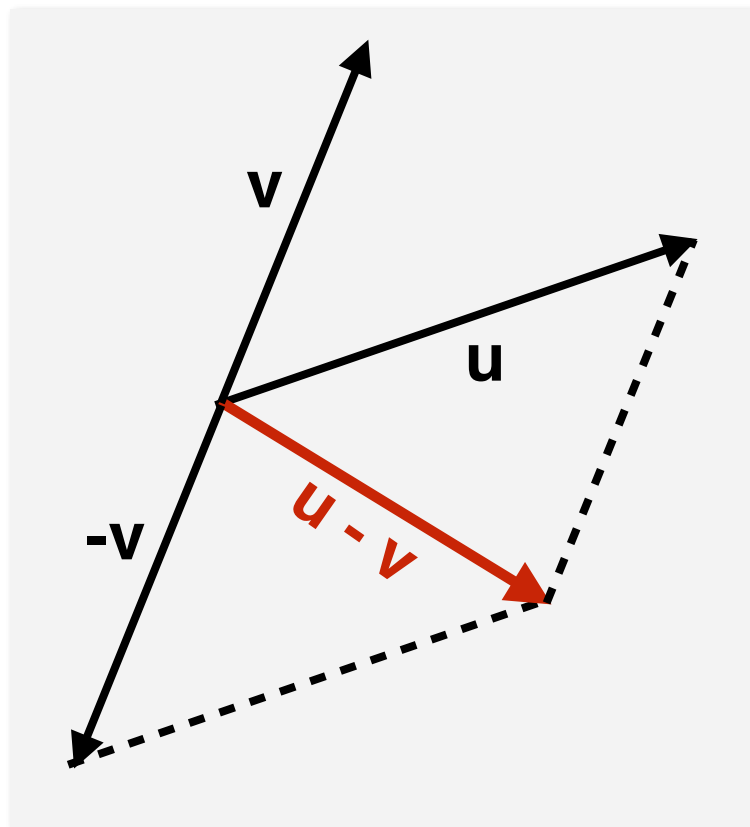
$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_x + v_x \\ u_y + v_y \end{bmatrix}$$



Multiplication by a scalar

$$\beta \mathbf{u} = \begin{bmatrix} \beta u_x \\ \beta u_y \end{bmatrix}$$

Vector operations



Subtraction

$$\mathbf{u} - \mathbf{v} = \begin{bmatrix} u_x - v_x \\ u_y - v_y \end{bmatrix}$$

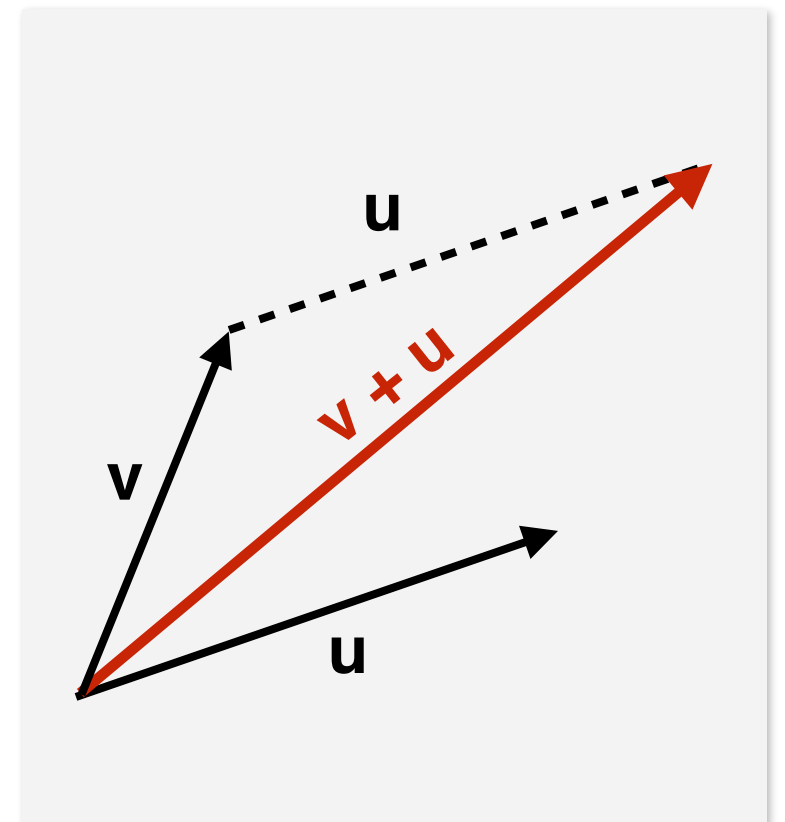
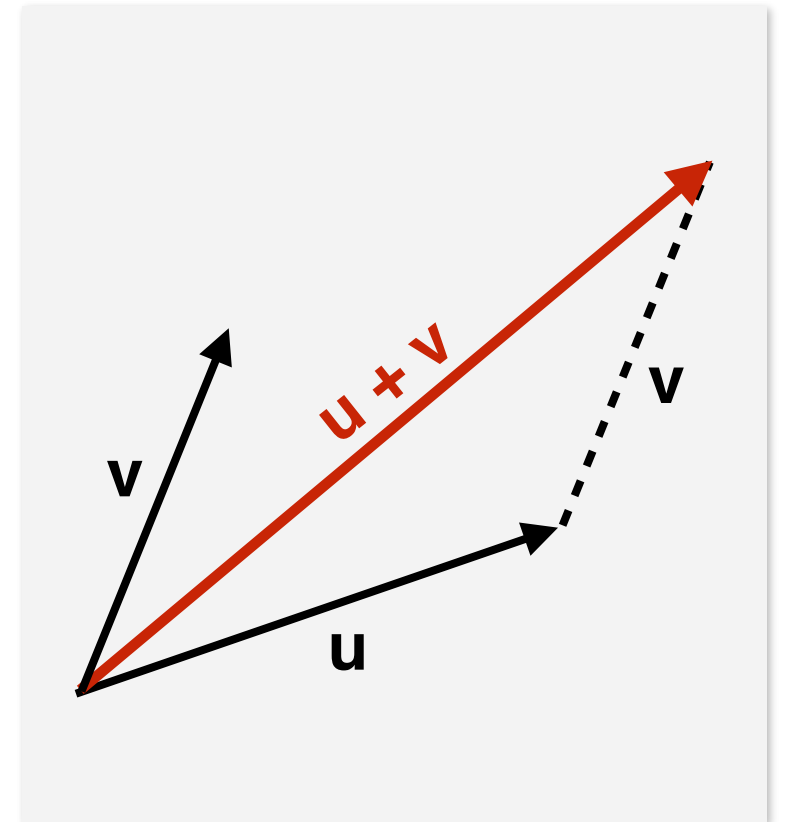
Properties of vector operations

- Addition - **associative** and **commutative**
- Scalar multiplication - **distributive** across addition

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$$

$$(k + p)\mathbf{u} = k\mathbf{u} + p\mathbf{u}$$



Vectors

- Two vectors are **parallels** if one is a scalar multiple of the other one

$$\mathbf{u} = k\mathbf{v}$$

- Parallel vectors - **linearly dependent**
- Non-parallel vectors - **linearly independent**

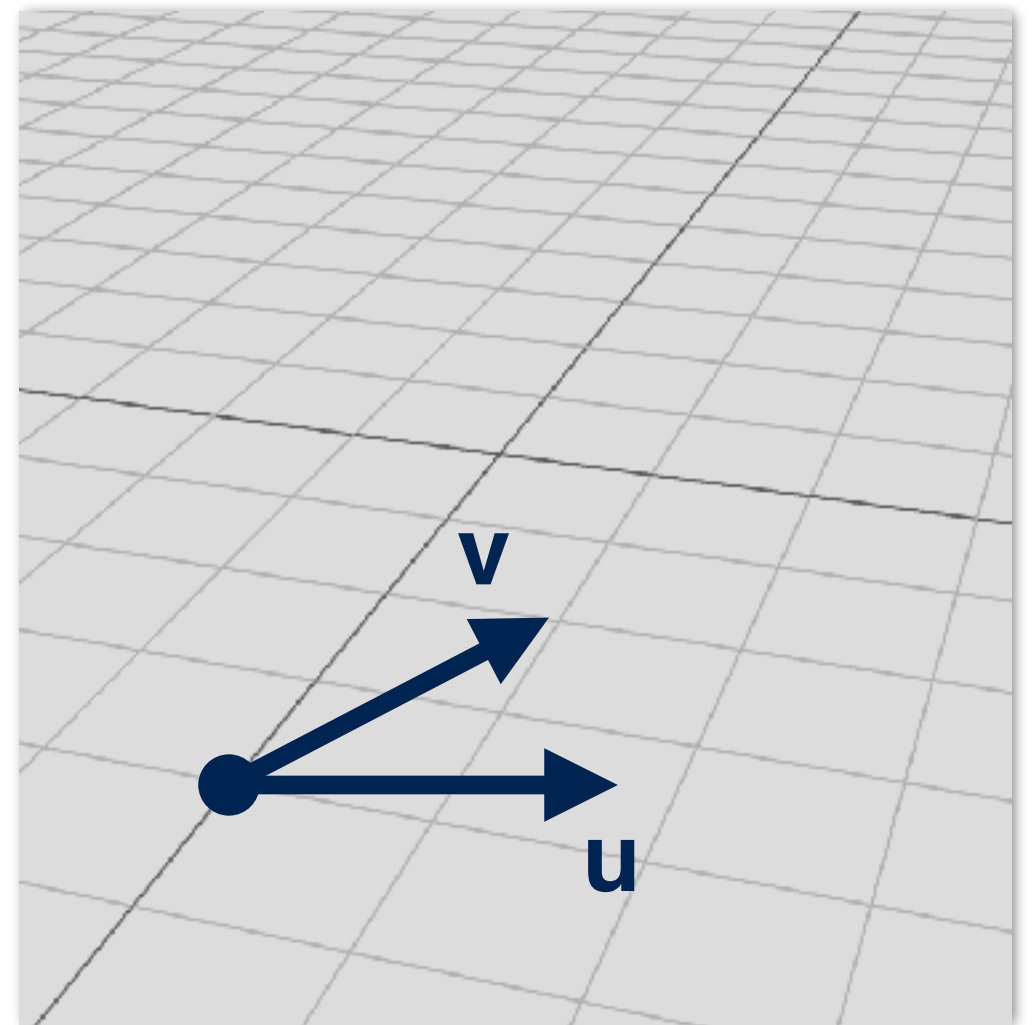
Span

- **Span** - the set of all vectors that can be defined as a linear combination of k vectors

$$\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k) = \{\mathbf{x} \in V \mid \mathbf{x} = \sum_{i=1}^k a_i \mathbf{u}_i, a_1, \dots, a_k \in \mathbb{R}\}$$

- ***Example:***

- span of two vectors $\mathbf{u}$ and $\mathbf{v}$ are all the vectors of the form $a_1 \mathbf{u} + a_2 \mathbf{v}$
for any two numbers a_1, a_2



Coordinate systems

- **Coordinate system** - the smallest set of vectors needed to produce the entire set of vectors using the vector operations
- **Basis** - used to describe **vectors**
- **Frame** - used to describe **points**
- Each coordinate system has a dimension and each of the basis vectors is called an axis
 - x axis, y axis, z axis - in 3D space

Basis and coordinate systems

- Any pair of linearly independent vectors form a **2D basis** (can be extended to higher dimensions)
- Two vectors form an **orthonormal basis** if:
 - they are orthogonal $\mathbf{x} \cdot \mathbf{y} = 0$
 - they are unit vectors $||\mathbf{x}|| = ||\mathbf{y}|| = 1$

Basis and coordinate systems

- In **3D**, the ***x***, ***y***, and ***z*** vectors form an orthonormal basis if:

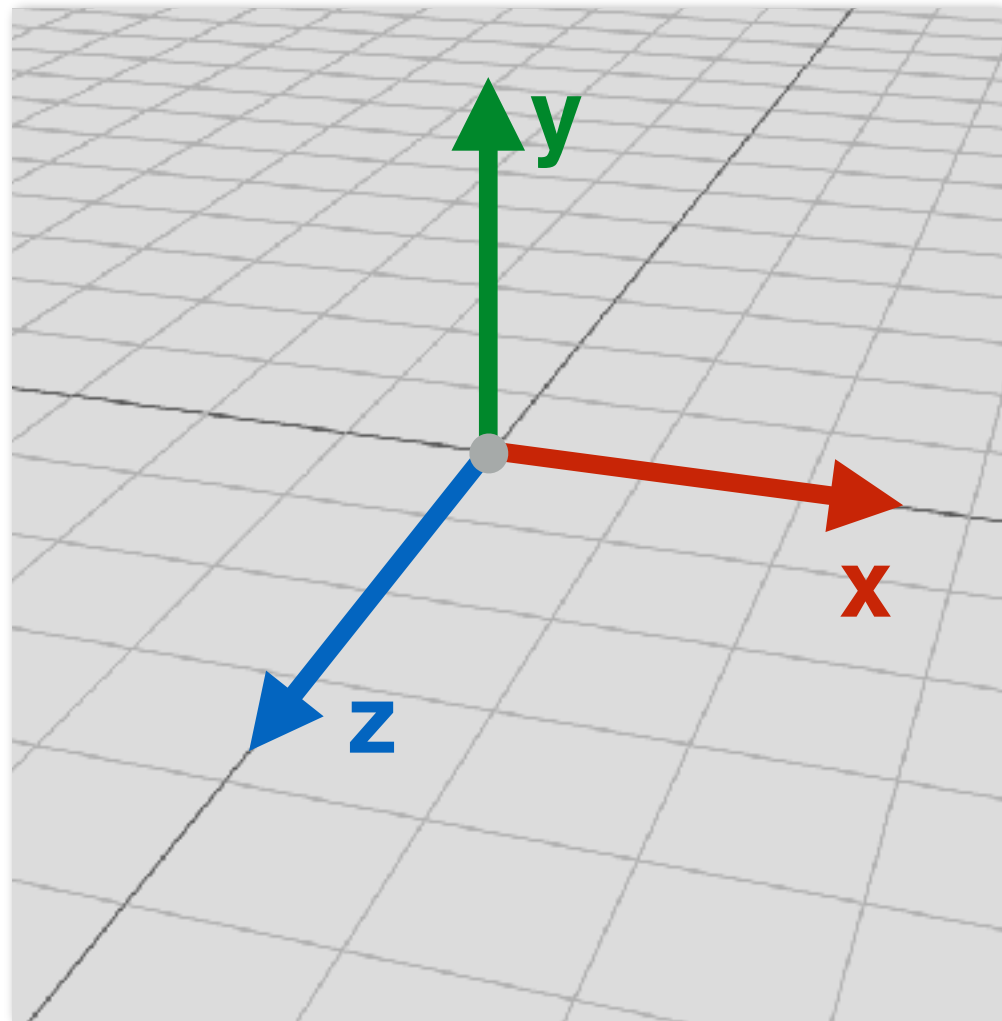
$$||\mathbf{x}|| = ||\mathbf{y}|| = ||\mathbf{z}|| = 1$$

$$\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{z} = \mathbf{z} \cdot \mathbf{x} = 0$$

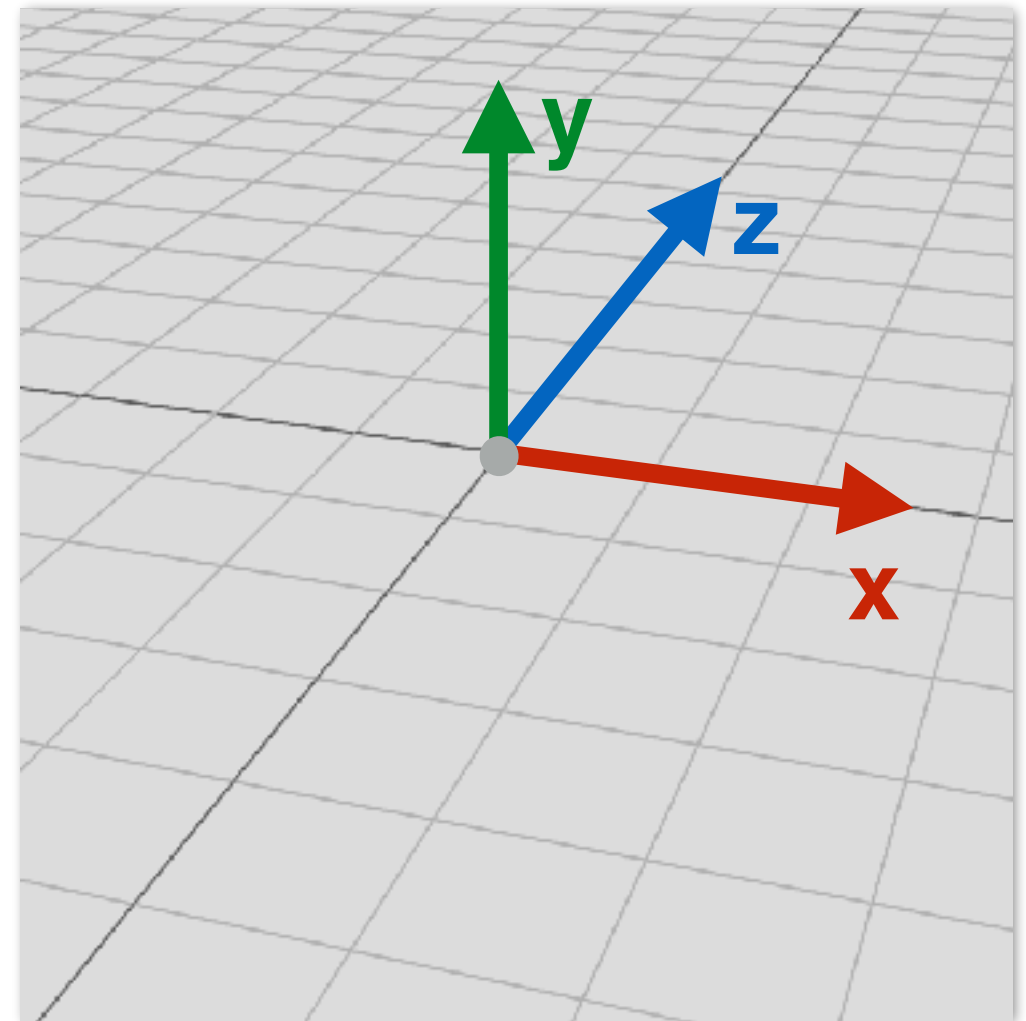
- The orthonormal basis is right-handed if:

$$\mathbf{z} = \mathbf{x} \times \mathbf{y}$$

Coordinate systems

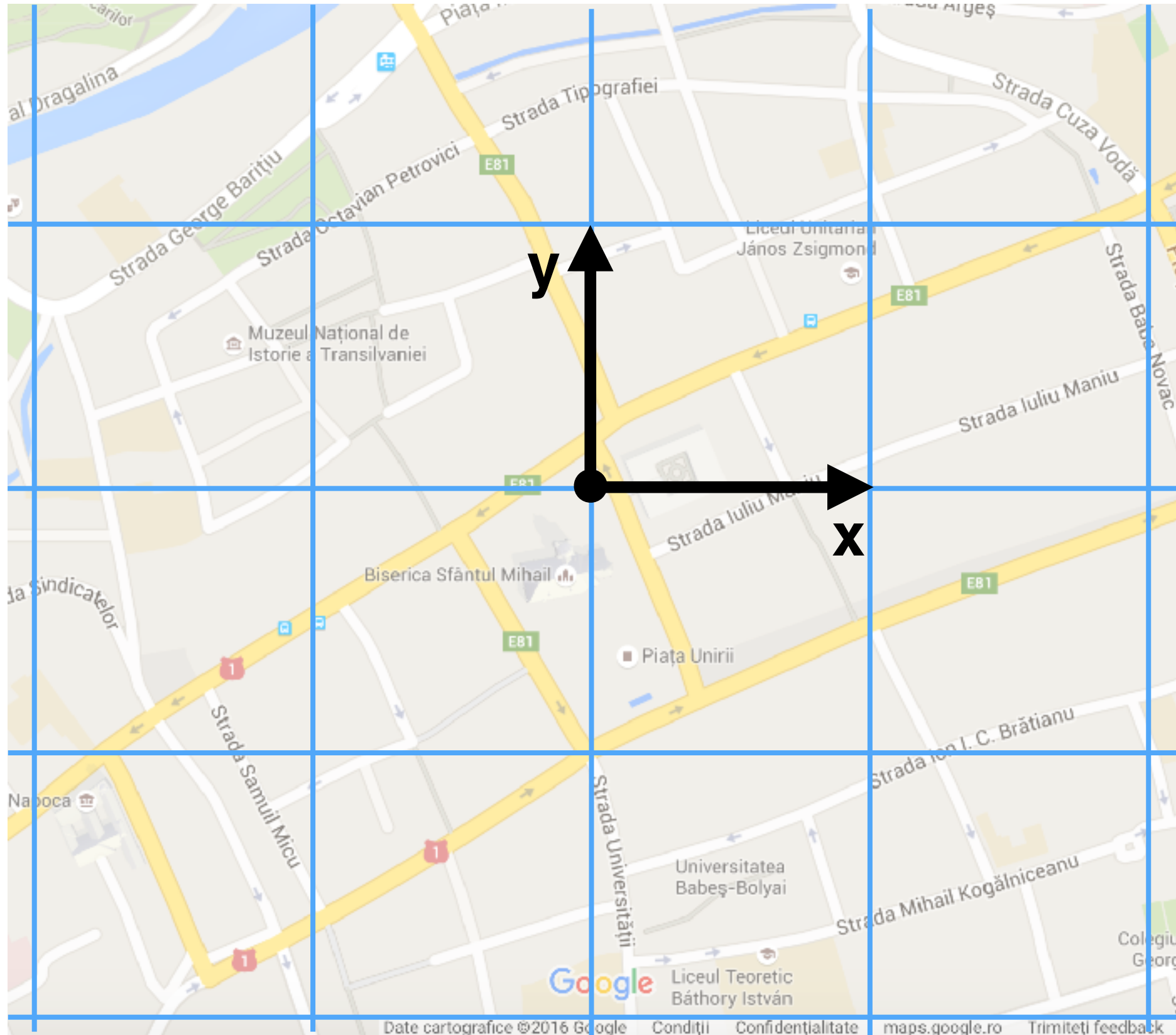


Right handed

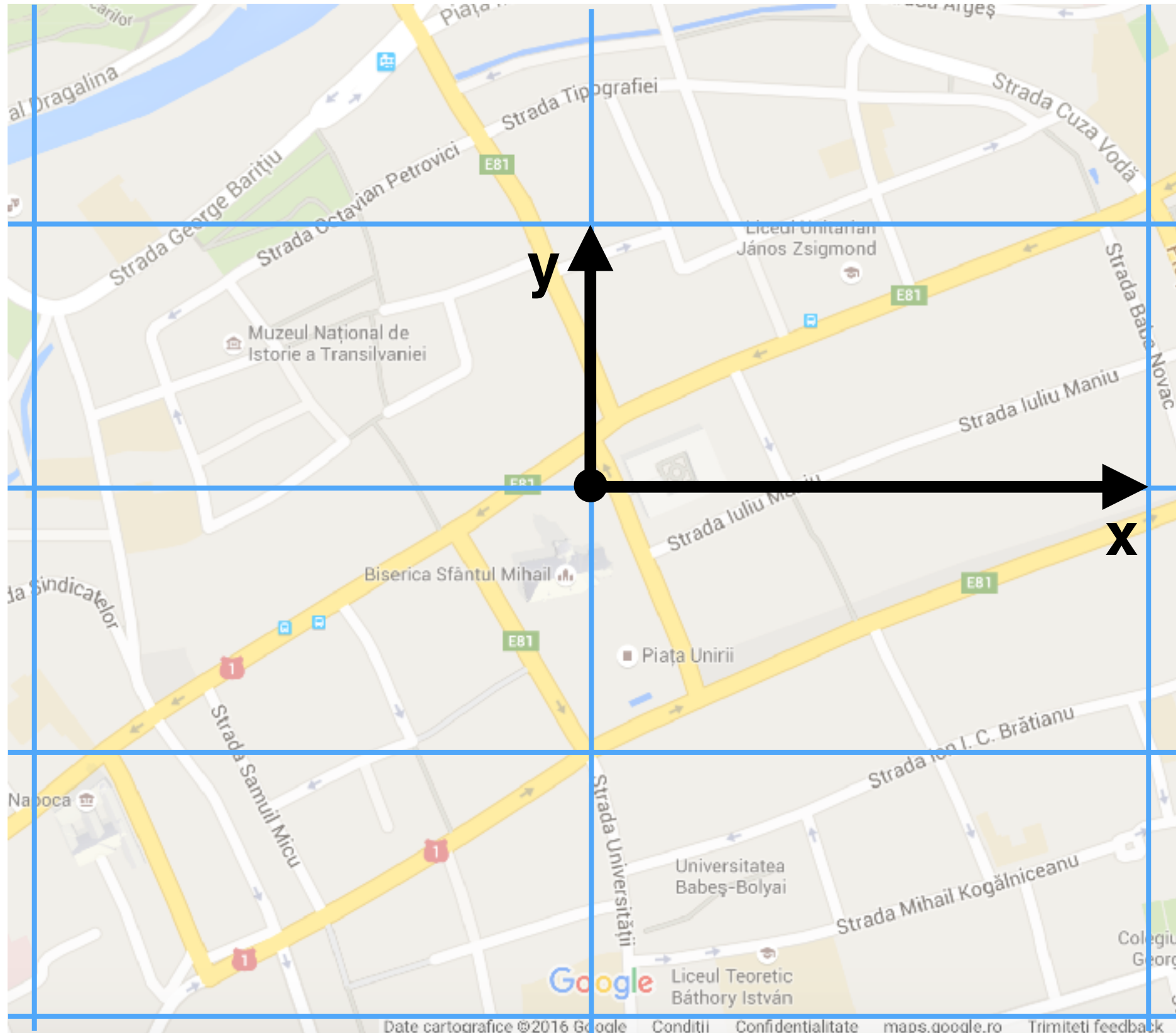


Left handed

Coordinate systems



Coordinate systems



Coordinate systems



Basis

- Any vector from the vector space can be expressed using a set of unique coordinates **$\mathbf{c}_i$**

$$\mathbf{v} = \sum_i c_i \mathbf{b}_i$$

Vectors

- In $\mathbf{R}^2$, any vector $[a \ b]^\top$ can be expressed:

$$\begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

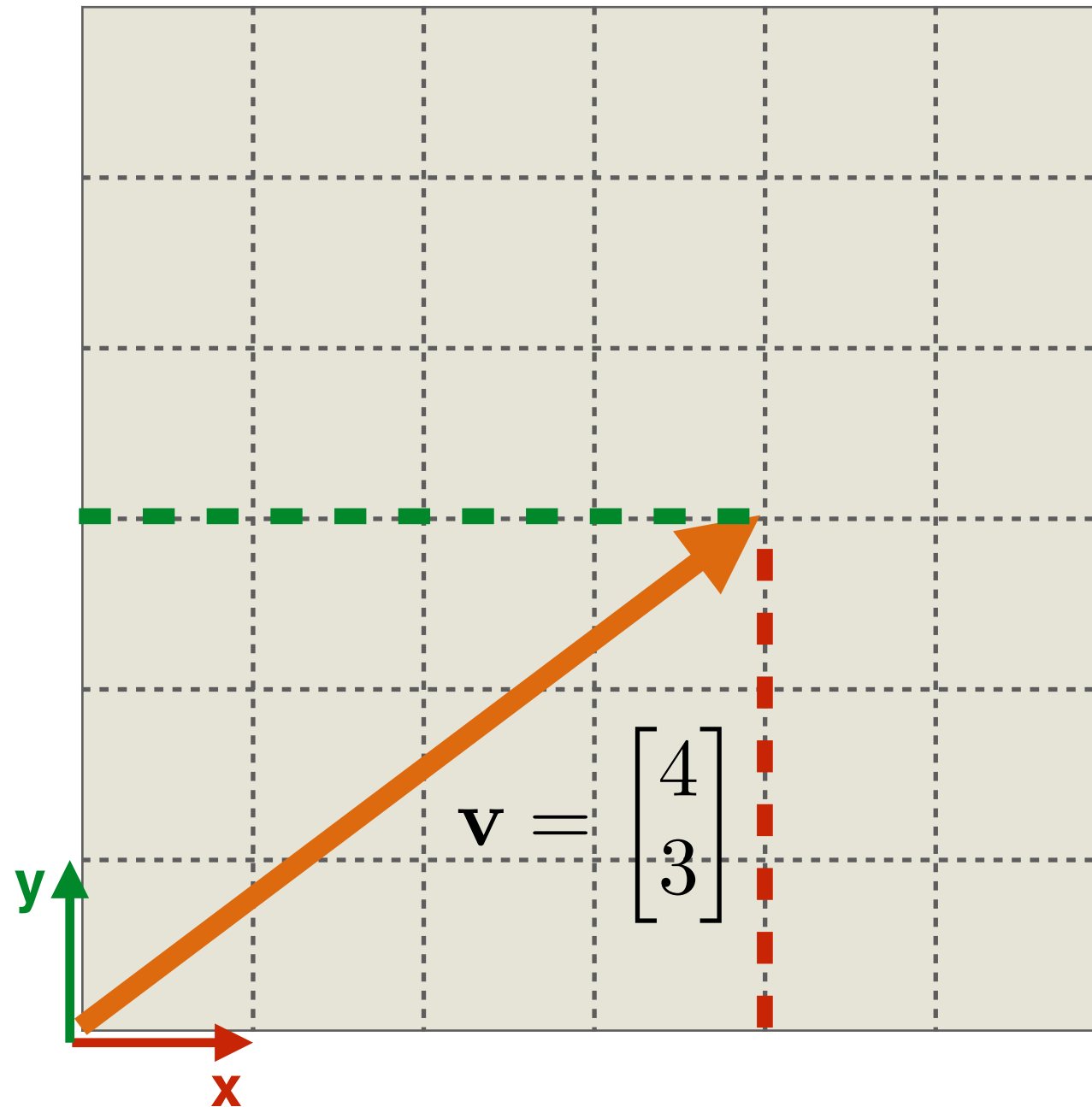
- In $\mathbf{R}^3$, we have a similar set of vectors to express any vector $[a \ b \ c]^\top$:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Coordinate vectors



Length of a vector

- Equals the square root of the sum of the squares of the elements
- For a 2D vector:

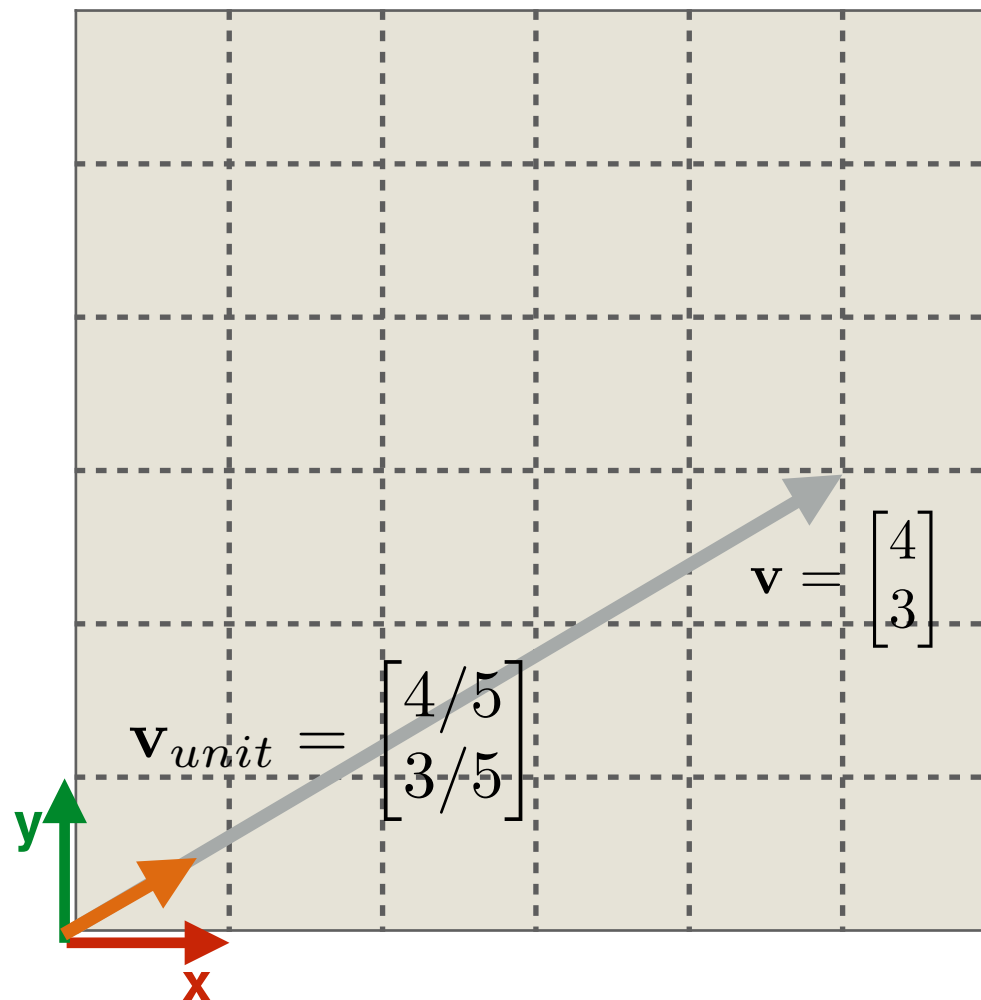
$$\text{length}(\mathbf{v}) = \|\mathbf{v}\| = \sqrt{x^2 + y^2}$$

- **Zero vector** $\|\mathbf{v}\| = 0$
- **Unit vector** $\|\mathbf{v}\| = 1$

Vector normalisation

- Used to convert a vector to a **unit vector**
- Done by dividing the vector by its length

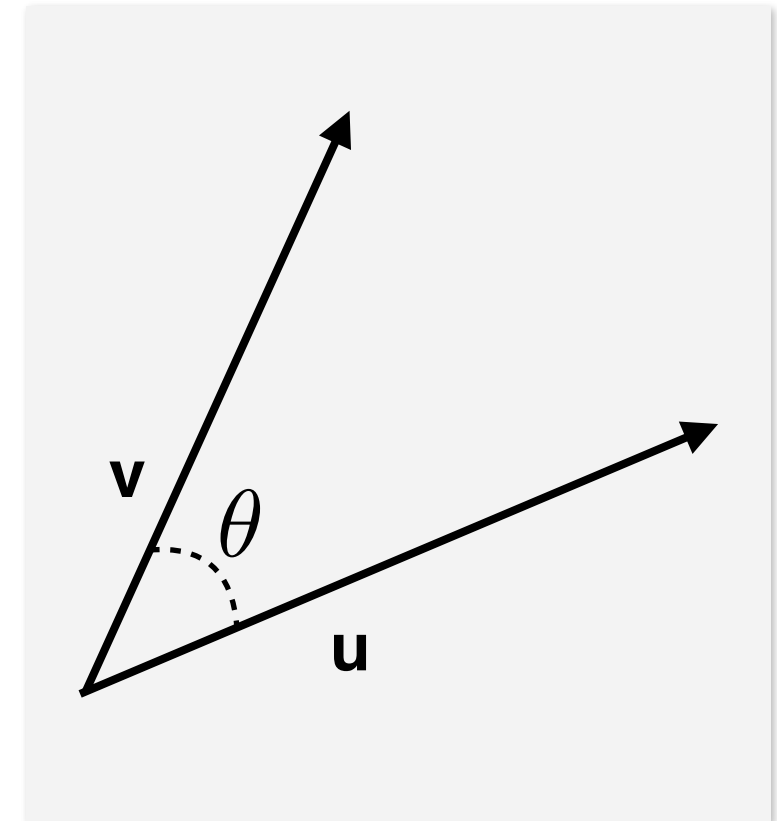
$$\mathbf{v}_{unit} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$



Dot product

- Also known as **scalar product**
- In **2D**: $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}_x \mathbf{v}_x + \mathbf{u}_y \mathbf{v}_y$
- In **3D**: $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}_x \mathbf{v}_x + \mathbf{u}_y \mathbf{v}_y + \mathbf{u}_z \mathbf{v}_z$
- Returns a value related to the vectors's length and the angle between them

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$



$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

Dot product

- **Associativity** and **distributivity** rules

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$$

$$\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$$

$$(k\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (k\mathbf{v}) = k\mathbf{u} \cdot \mathbf{v}$$

- In a **Cartesian** coordinate system:

$$\mathbf{x} \cdot \mathbf{x} = \mathbf{y} \cdot \mathbf{y} = 1$$

$$\mathbf{x} \cdot \mathbf{y} = 0$$

Uses of dot product

- Compute the **length** of a vector

$$\text{length}(\mathbf{v}) = \|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

- Compute the **distance** between two points P and Q

$$\mathbf{v} = Q - P, \text{distance}(P, Q) = \text{length}(\mathbf{v}) = \|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

Uses of dot product

- Checking two vectors for **orthogonality**

$$\mathbf{u} \cdot \mathbf{v} = 0$$

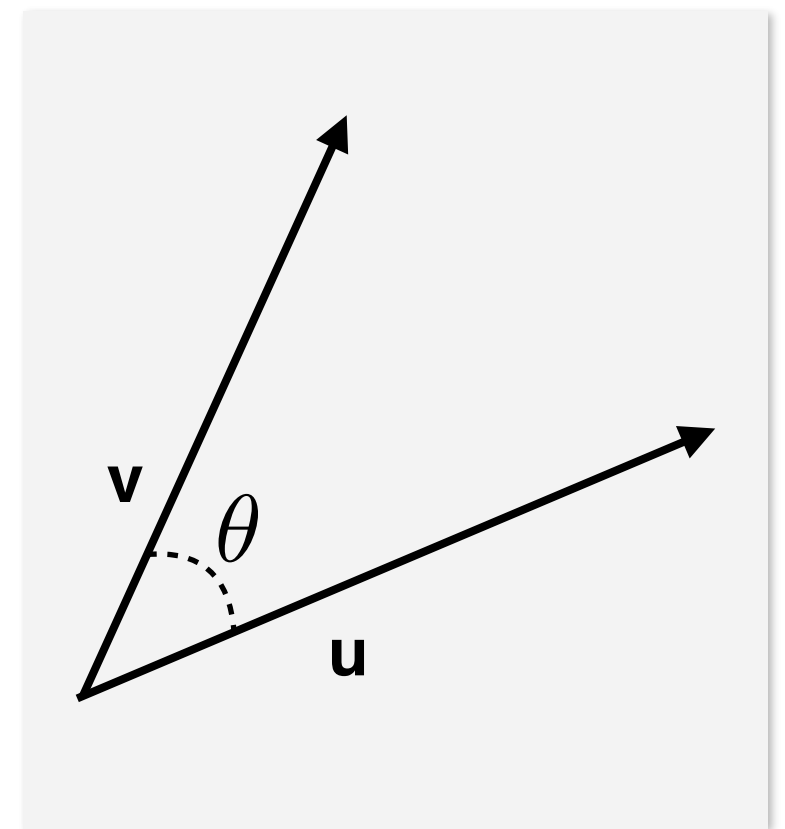
- Compute the **angle** between two vectors

$$\theta = \cos^{-1} \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

$$\mathbf{u} \cdot \mathbf{v} < 0 \Rightarrow \theta > 90^\circ$$

$$\mathbf{u} \cdot \mathbf{v} = 0 \Rightarrow \theta = 90^\circ$$

$$\mathbf{u} \cdot \mathbf{v} > 0 \Rightarrow \theta < 90^\circ$$

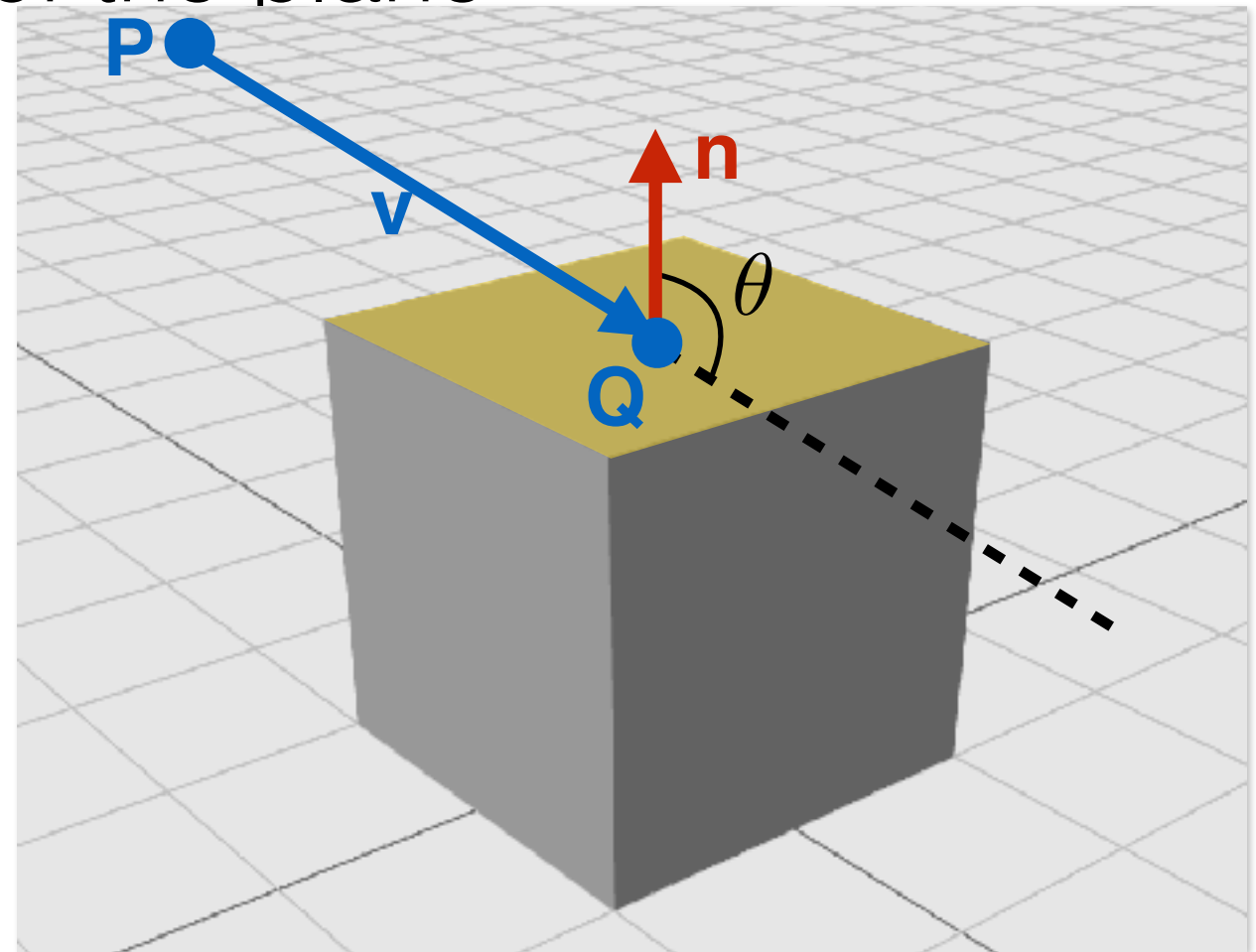


Uses of dot product

Relative position of a point P against a plane

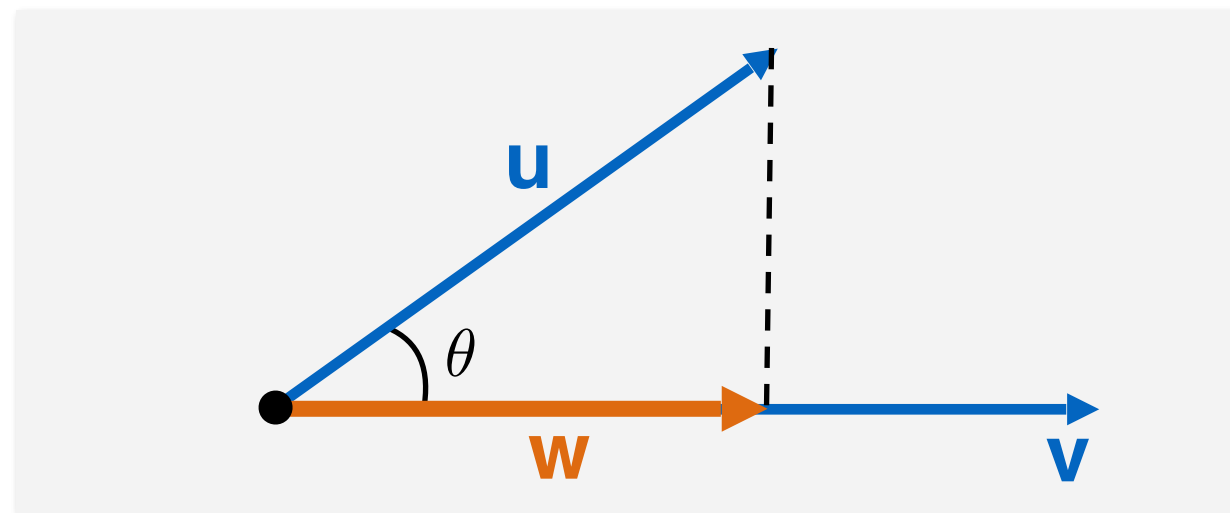
- $\theta > 90^\circ$ then P is in front of the plane
- $\theta = 90^\circ$ then P is on the plane
- $\theta < 90^\circ$ then P is on back of the plane

If the viewer (P) is in front of the polygon of a convex 3D object, the polygon is **visible**



Uses of dot product

Projecting a vector onto another



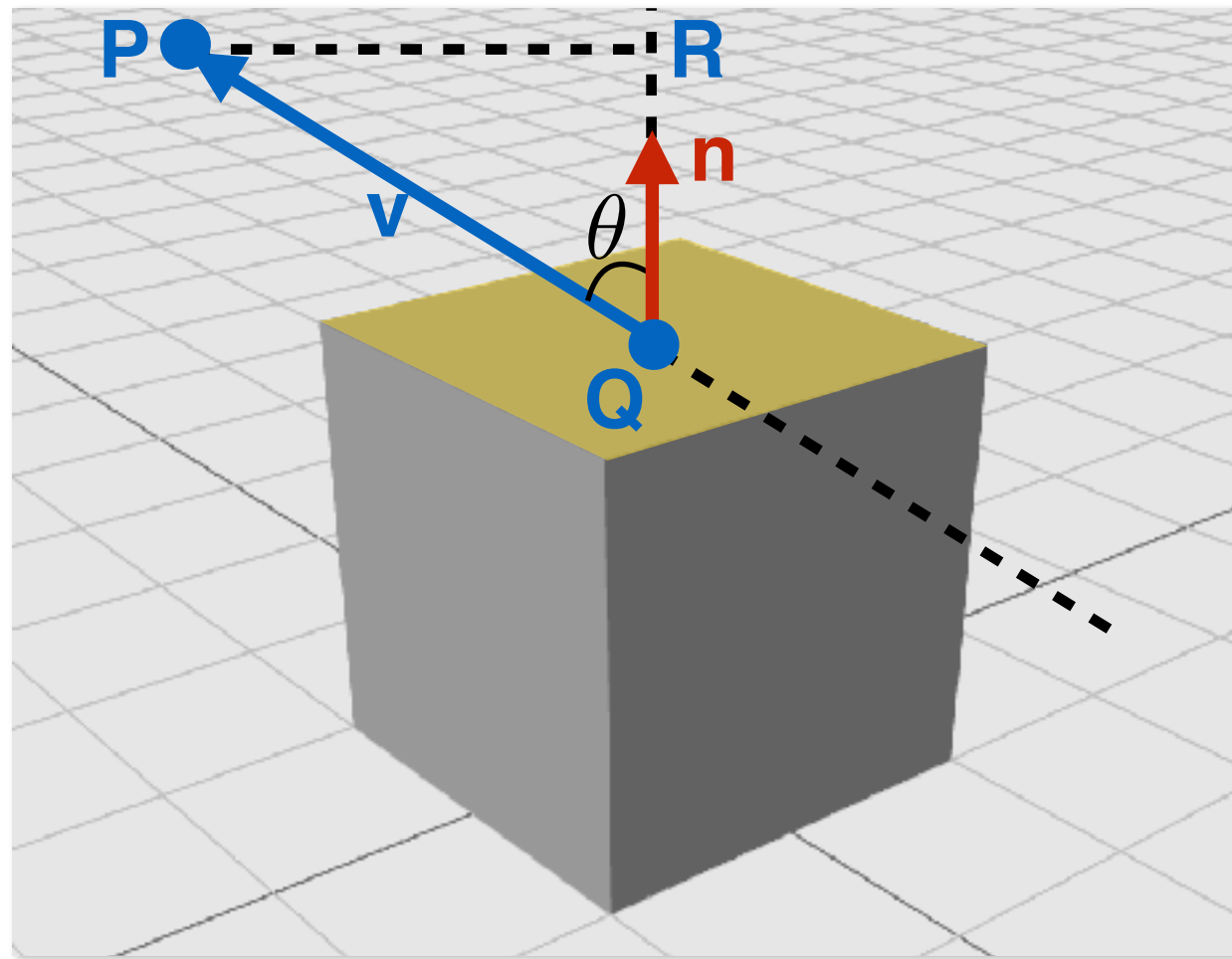
$$\|\mathbf{v}\| = 1$$

$$\|\mathbf{w}\| = \|\mathbf{u}\| \cos \theta = \|\mathbf{u}\| \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \mathbf{u} \cdot \mathbf{v}$$

Uses of dot product

Compute the distance between point P and a plane

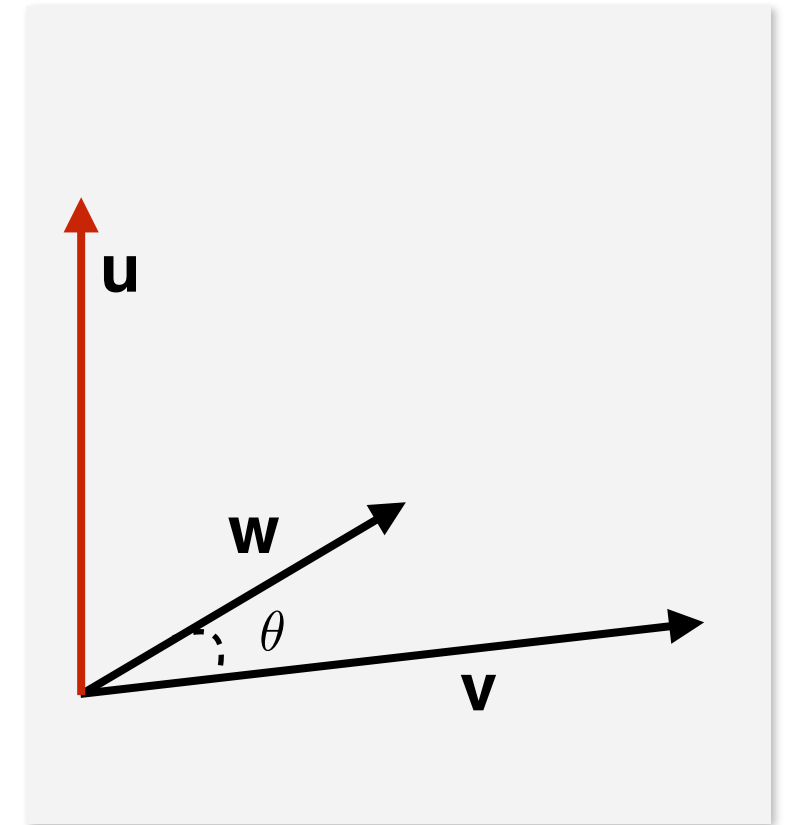
$$\|\mathbf{w}\| = \|\mathbf{v}\| \cos \theta = \|\mathbf{v}\| \frac{\mathbf{n} \cdot \mathbf{v}}{\|\mathbf{n}\| \|\mathbf{v}\|} = \frac{\mathbf{n} \cdot \mathbf{v}}{\|\mathbf{n}\|}$$



Cross product

- The cross product or vector product of two vectors is a vector orthogonal to both
- Right-hand rule dictates direction of cross product
- The length of the resulting vector:

$$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \|\mathbf{w}\| \sin \theta$$



$$\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \times \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} = \begin{bmatrix} v_y w_z - v_z w_y \\ v_z w_x - v_x w_z \\ v_x w_y - v_y w_x \end{bmatrix}$$

Cross product

$$\mathbf{u} = \mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

$$\mathbf{u} = \mathbf{e}_1 \begin{vmatrix} v_y & v_z \\ w_y & w_z \end{vmatrix} - \mathbf{e}_2 \begin{vmatrix} v_x & v_z \\ w_x & w_z \end{vmatrix} + \mathbf{e}_3 \begin{vmatrix} v_x & v_y \\ w_x & w_y \end{vmatrix}$$

$$\mathbf{u} = (v_y w_z - v_z w_y) \mathbf{e}_1 + (v_z w_x - v_x w_z) \mathbf{e}_2 + (v_x w_y - v_y w_x) \mathbf{e}_3$$

$$\mathbf{u} = (v_y w_z - v_z w_y) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (v_z w_x - v_x w_z) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + (v_x w_y - v_y w_x) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} v_y w_z - v_z w_y \\ v_z w_x - v_x w_z \\ v_x w_y - v_y w_x \end{bmatrix}$$

Cross product properties

- **Anticommutative:**

$$\mathbf{v} \times \mathbf{w} = -\mathbf{w} \times \mathbf{v}$$

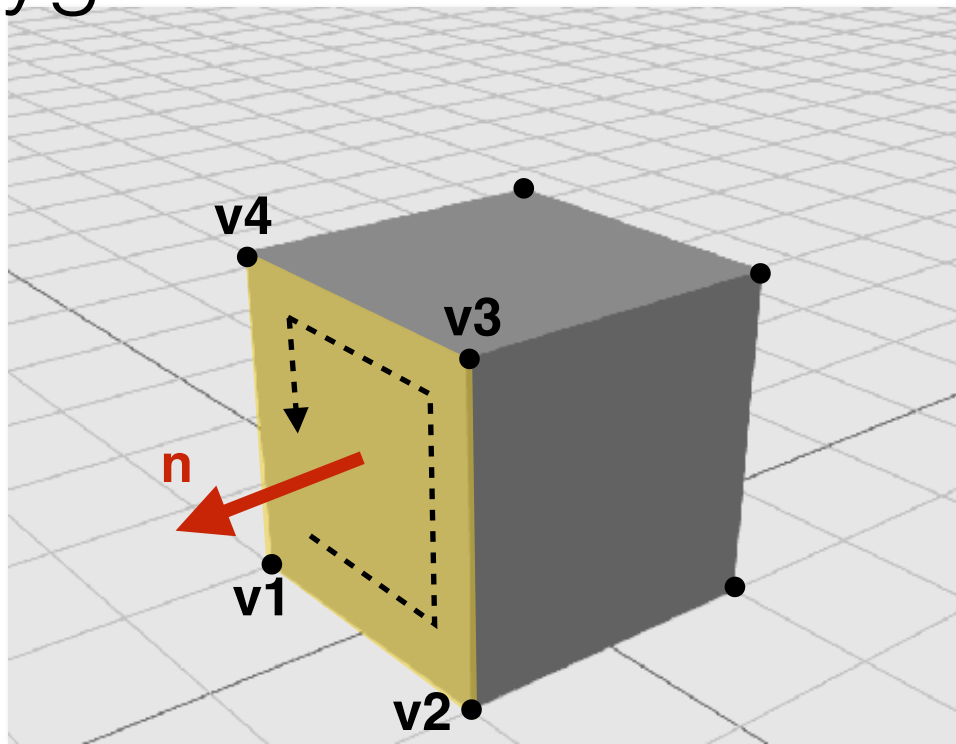
- **Distributive** over addition and scalar multiplication:

$$\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$$

$$\mathbf{a} \times (k\mathbf{b}) = k(\mathbf{a} \times \mathbf{b})$$

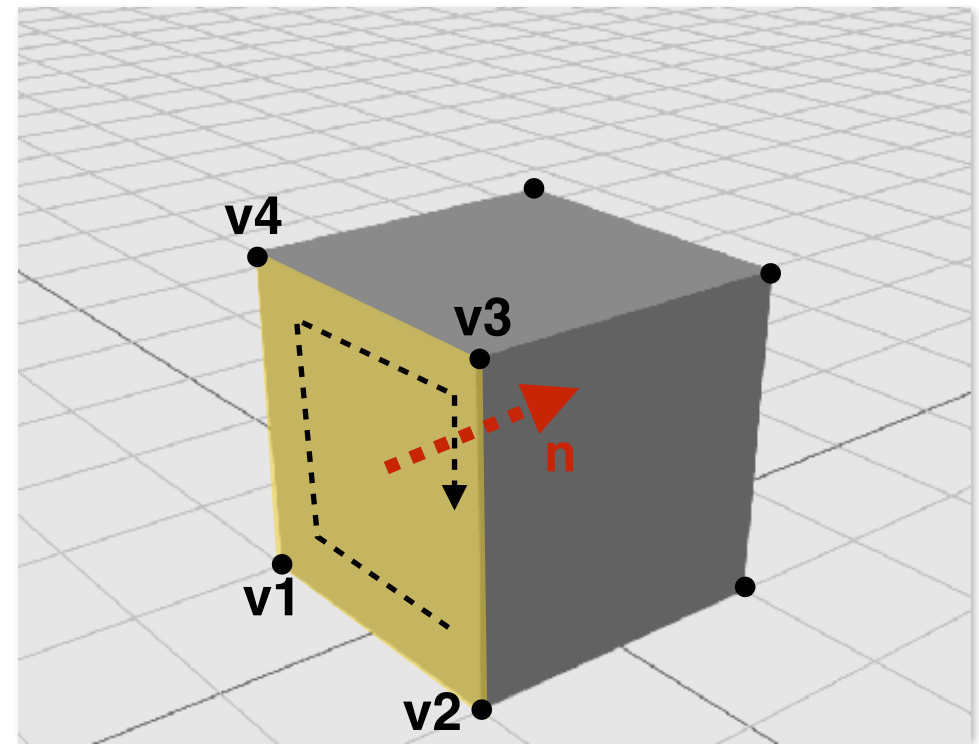
Uses of cross product

Cross product computes the **normal vector** ($\mathbf{n} = \mathbf{e}_1 \times \mathbf{e}_2$), where $\mathbf{e}_1$ and $\mathbf{e}_2$ are two consecutive edges of a polygon



Normal vector points **away** of the object

- **Counterclockwise** vertex sequence, e.g. v_1, v_2, v_3, v_4

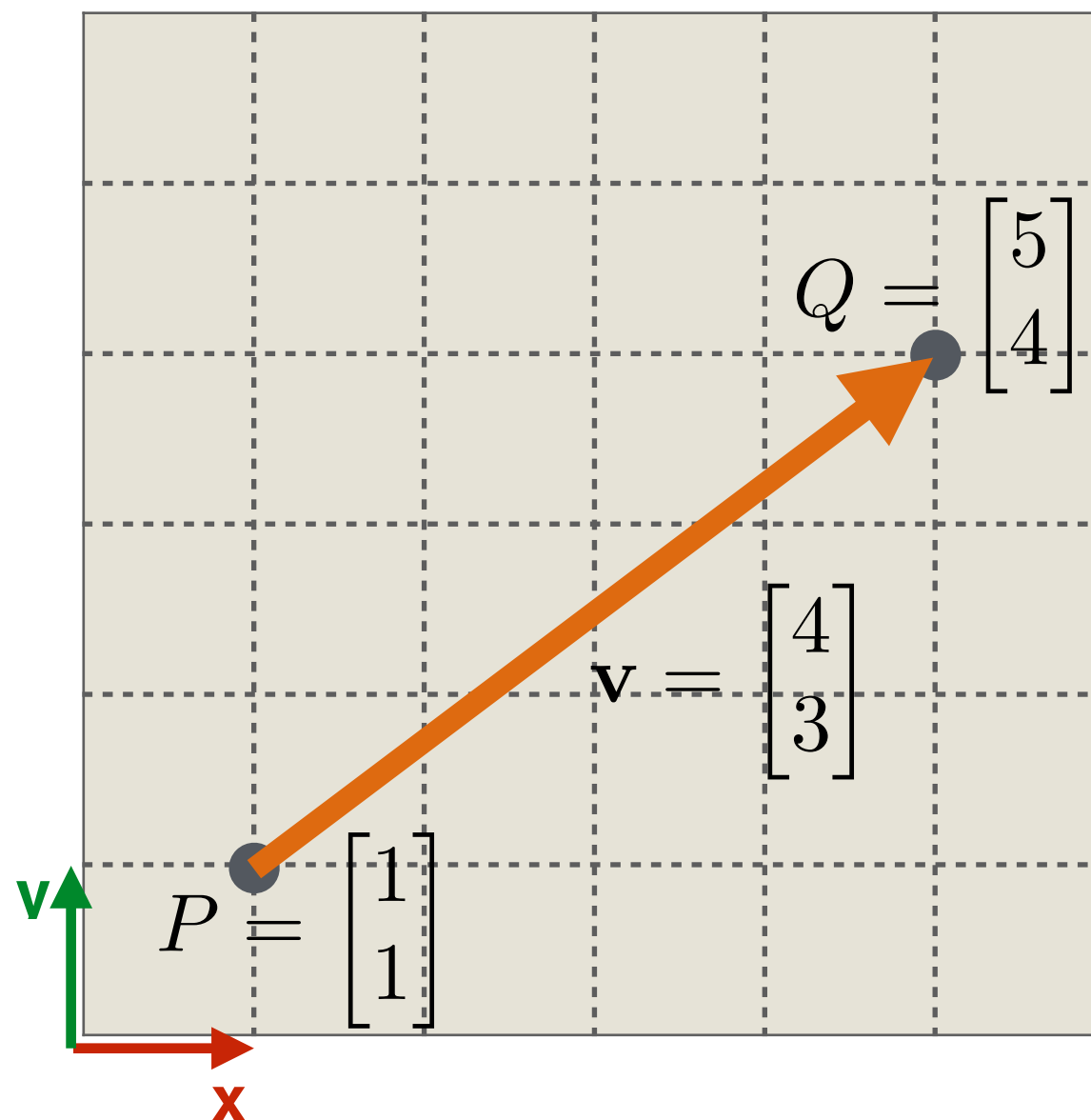


Normal vector points **into** the object

- **Clockwise** vertex sequence, e.g. v_1, v_4, v_3, v_2

Points vs Vectors

- **Points** represent **places** (locations)
- **Vectors** represent **displacement** (motion) needed to move from point to point



Operations on Points and Vectors

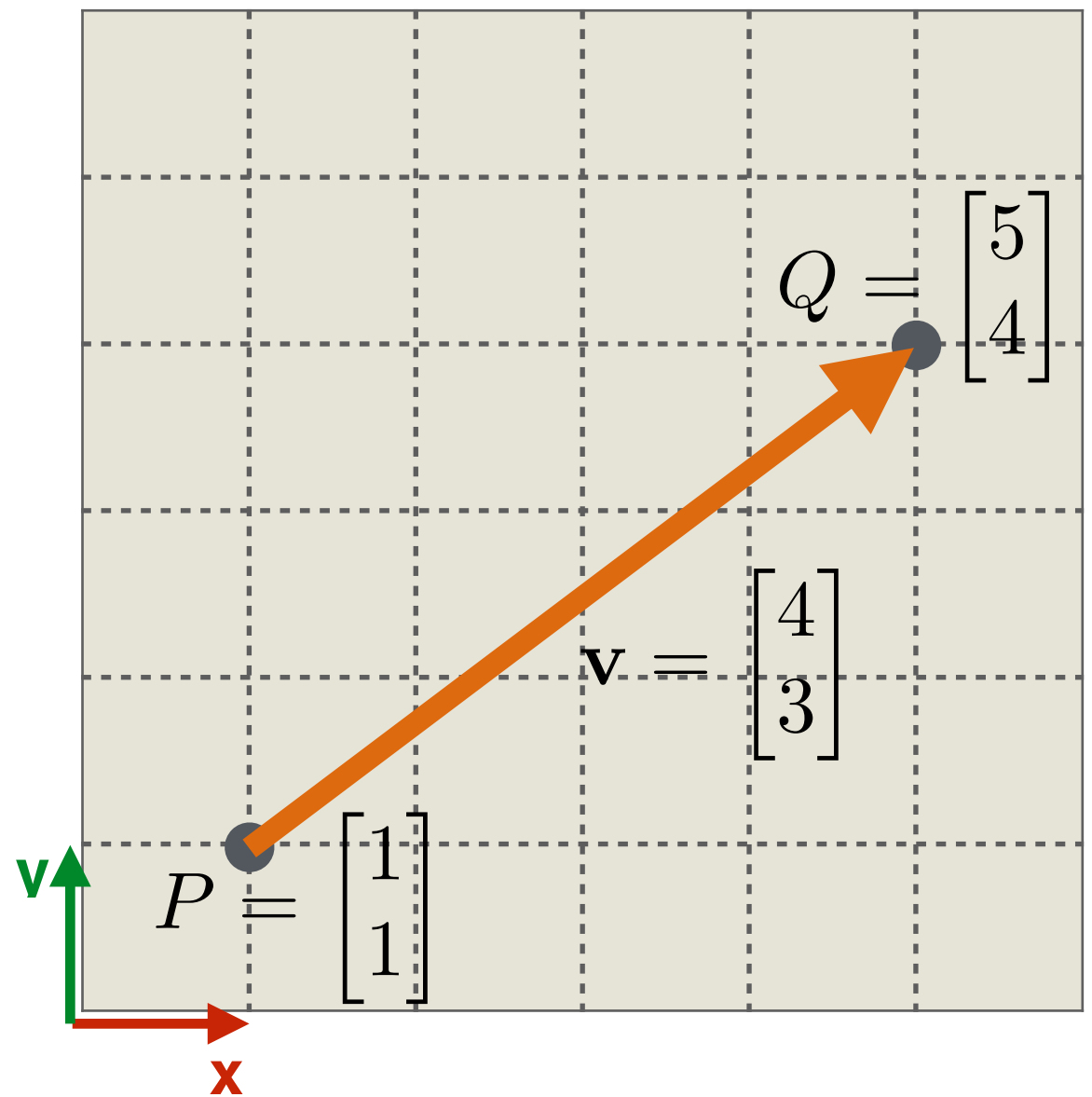
- Point-point **subtraction** (results in a vector pointing from P to Q)

$$Q - P = \mathbf{v}$$

- Vector-point **addition** (results a new point)

$$P + \mathbf{v} = Q$$

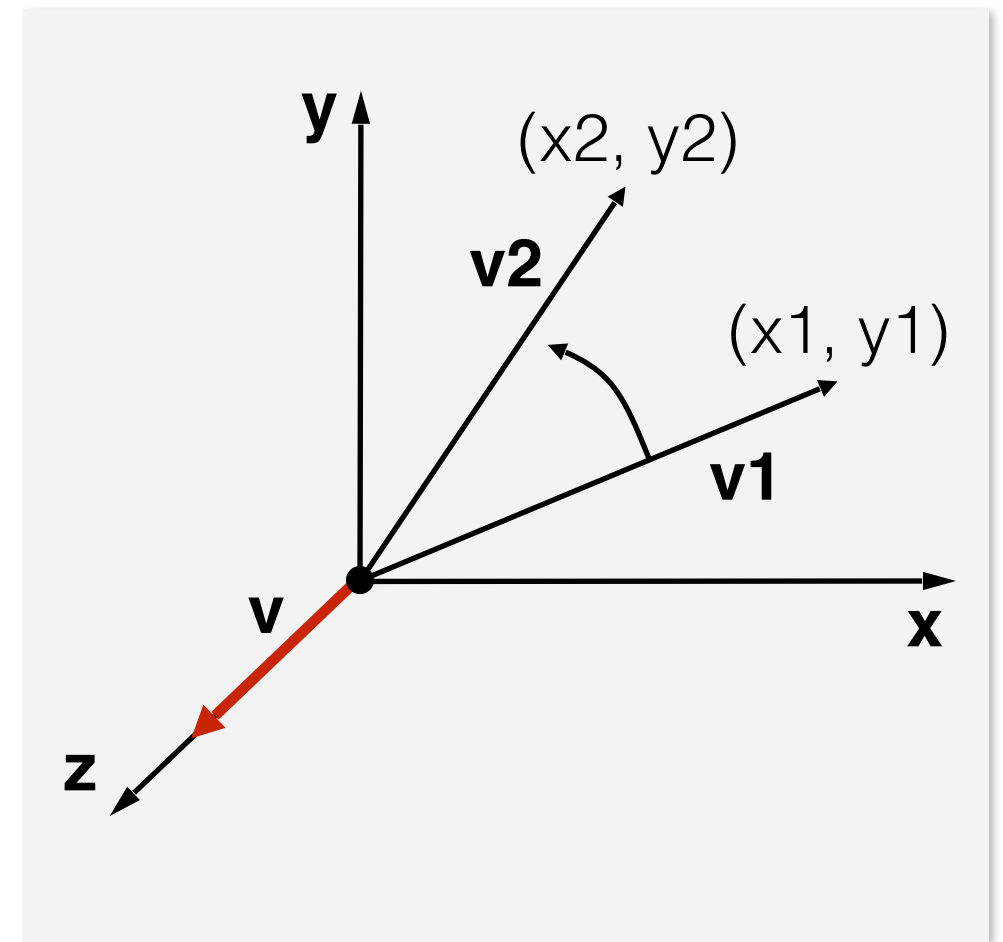
- The sum of two points isn't defined at all (? Cluj+București?)



Area of a triangle

$$\mathbf{v} = \mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ x_1 & y_1 & 0 \\ x_2 & y_2 & 0 \end{vmatrix}$$

$$\mathbf{v} = (x_1 y_2 - x_2 y_1) \mathbf{e}_3$$



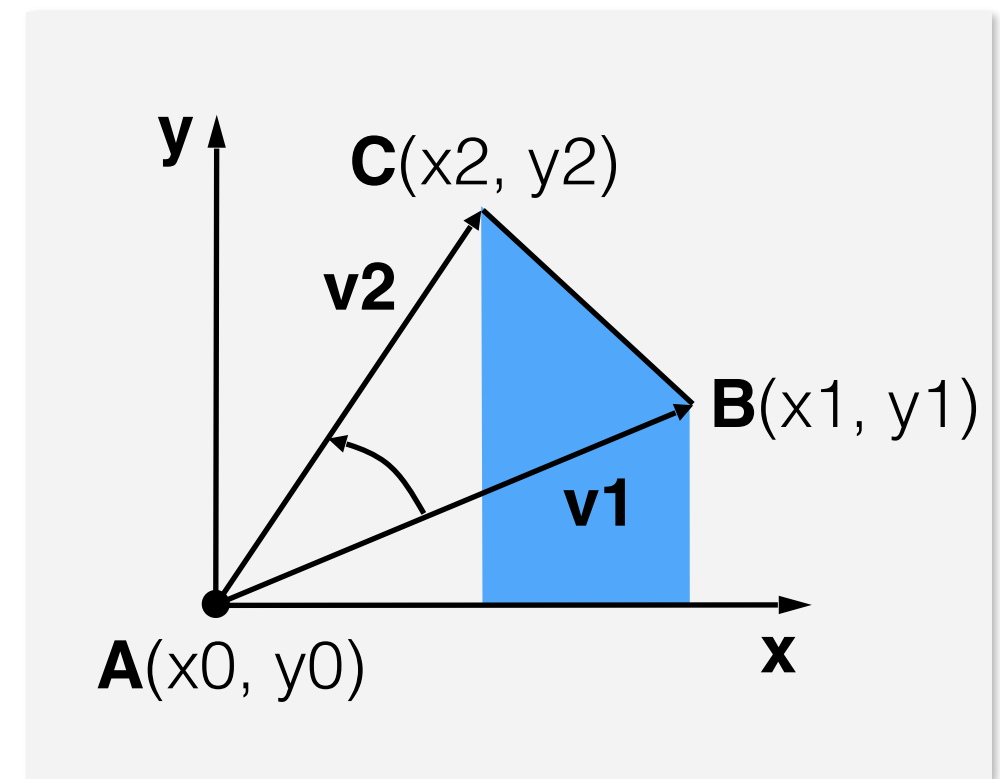
Area of a triangle

- The triangle ABC has the area:

$$\frac{1}{2}(x_1 - x_2)(y_1 + y_2) + \frac{1}{2}(x_2 - x_0)(y_2 + y_0) + \frac{1}{2}(x_0 - x_1)(y_0 + y_1)$$

$$\frac{1}{2}(x_0y_1 - x_1y_0 + x_1y_2 - x_2y_1 + x_2y_0 - x_0y_2)$$

$$S = \frac{1}{2}(x_1y_2 - x_2y_1) = \frac{1}{2}\|\mathbf{v}_1 \times \mathbf{v}_2\|$$



Area of a polygon

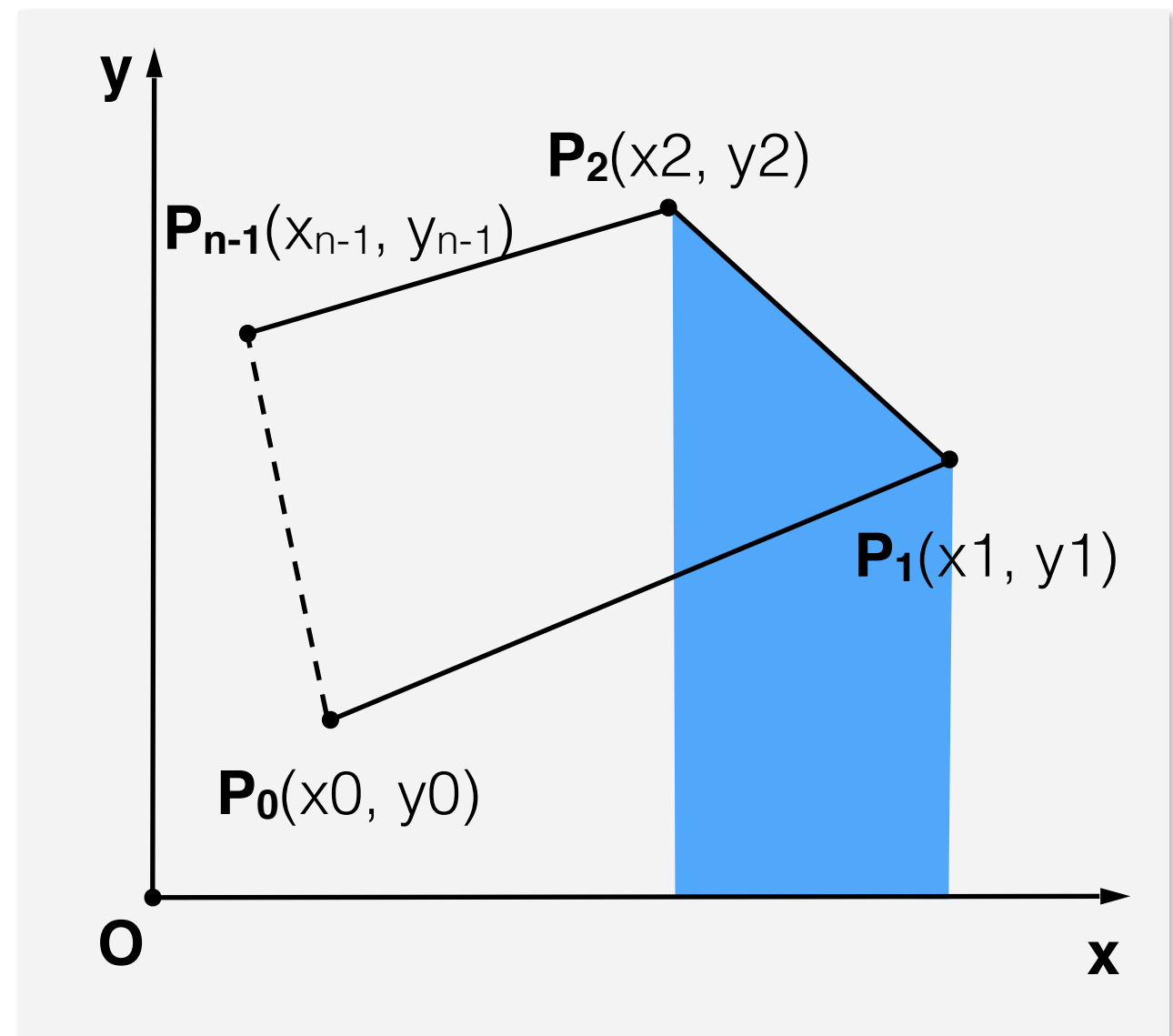
- Polygonal area:

$$\frac{1}{2} (x_0 - x_1) (y_0 + y_1) + \frac{1}{2} (x_1 - x_2) (y_1 + y_2) + \dots + \frac{1}{2} (x_n - x_0) (y_n + y_0) =$$

$$\frac{1}{2} (x_0 y_0 + x_0 y_1 - x_1 y_0 - x_1 y_1 + x_1 y_1 + x_1 y_2 - x_2 y_1 - x_2 y_2 + \dots + x_n y_n + x_n y_0 - x_0 y_n - x_0 y_0)$$

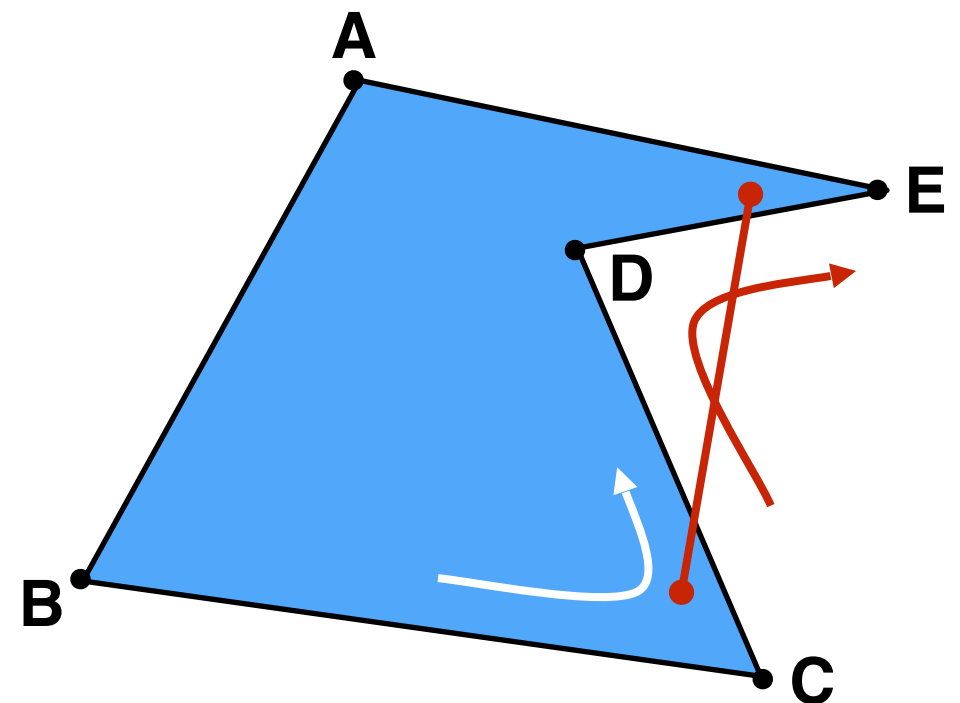
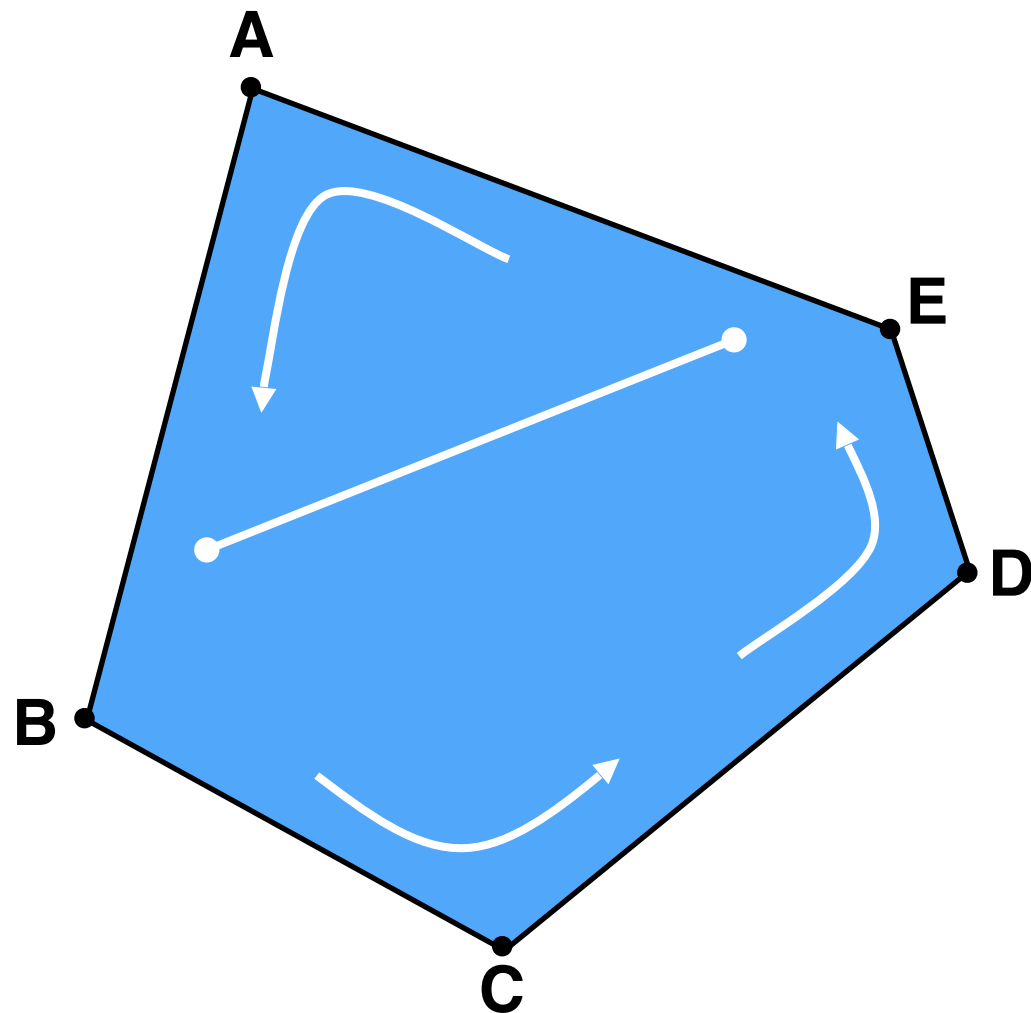
$$S = \frac{1}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i)$$

$$S = \frac{1}{2} \sum_{i=0}^{n-1} \|\mathbf{v}_i \times \mathbf{v}_{i+1}\|$$



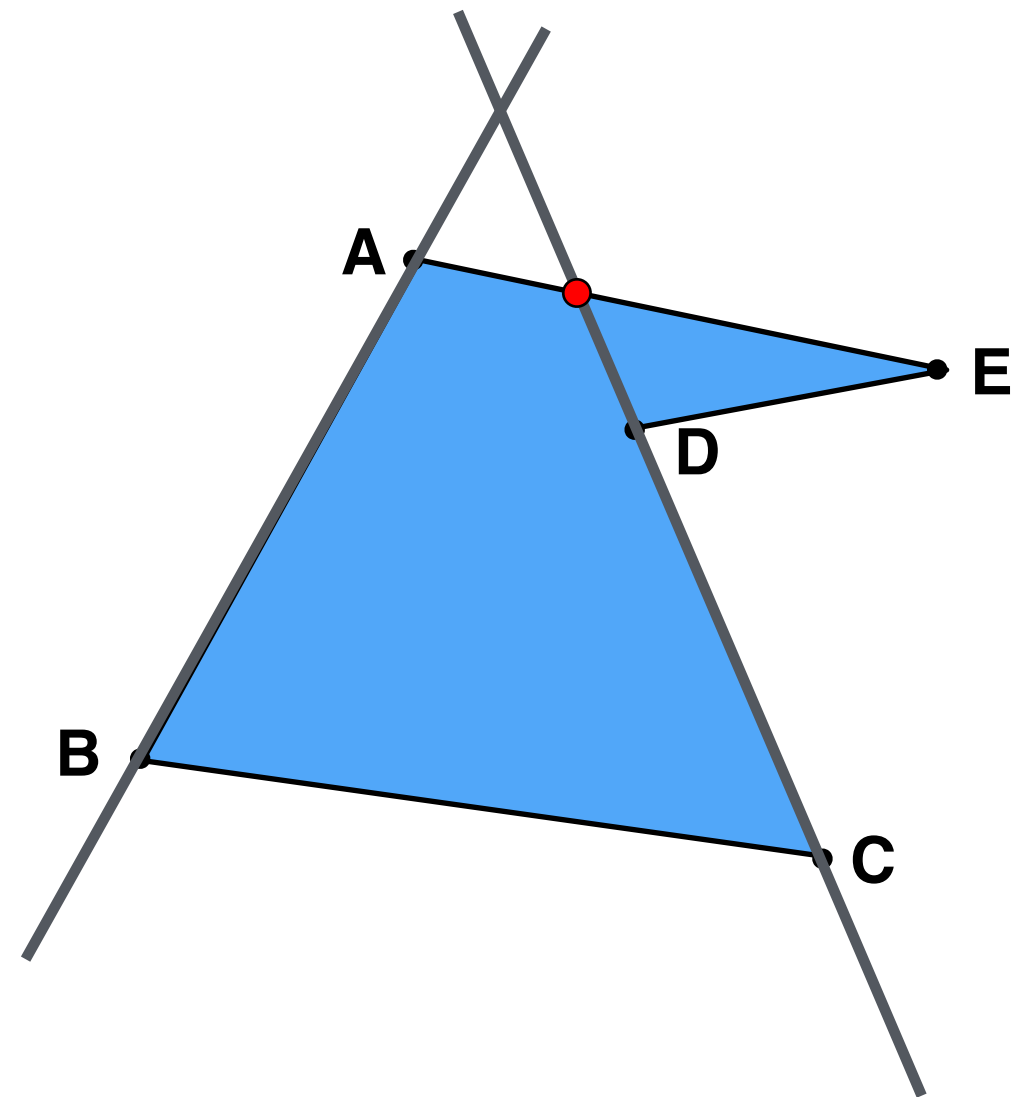
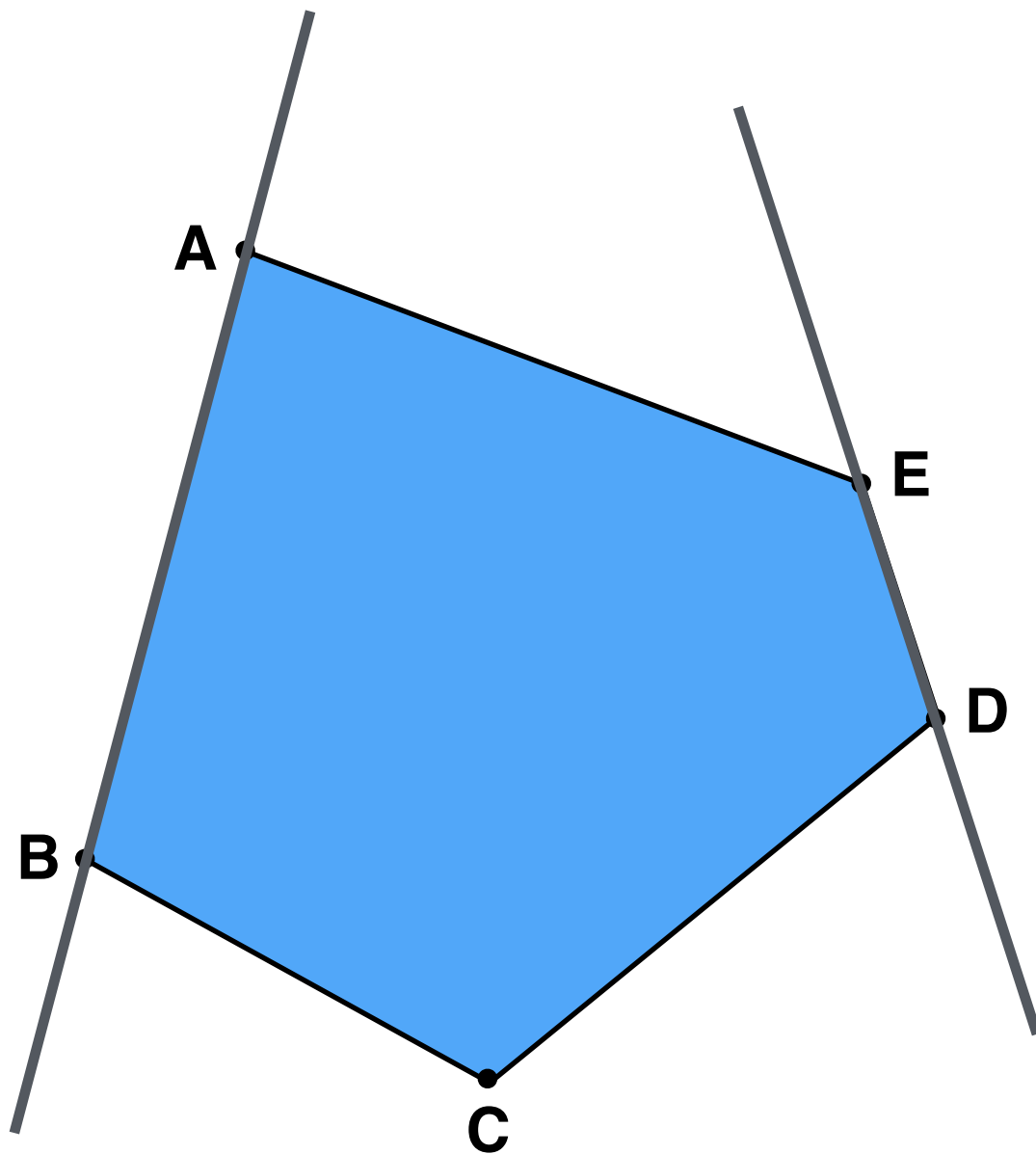
Convex and concave polygon

- Cross product of successive edges
- Turn directions
- Order of three successive vertices
- Angles
- Intersection between an edge and the polygon



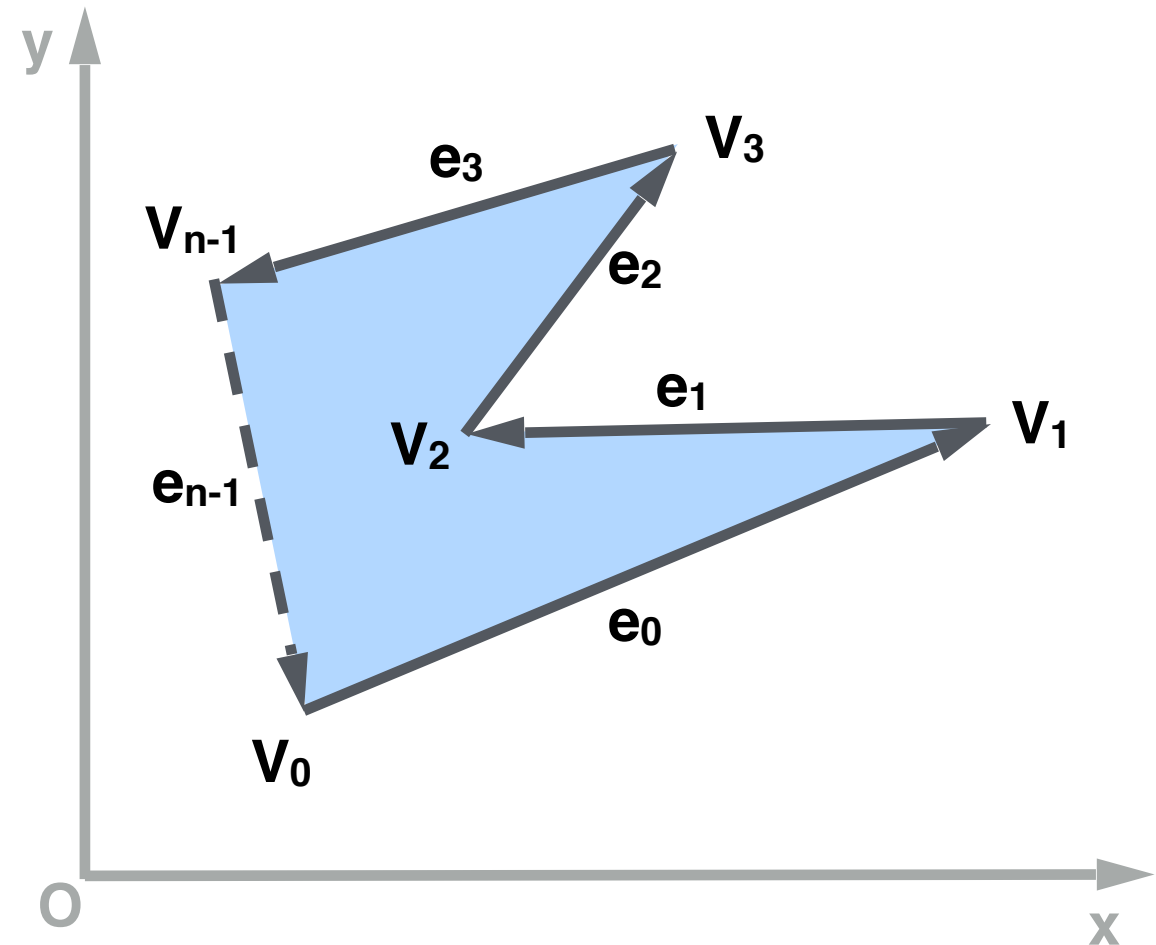
Convex and concave polygon

- At least one intersection between an edge and the polygon



Test for convex and concave 2D polygons

- Transform polygon to **XY plane**
- Compute edge vectors in counterclockwise order:
$$\mathbf{e}_k = \mathbf{V}_{k+1} - \mathbf{V}_k$$
- Set for each edge vector the **z** component to **0**
- For each pair of consecutive edge vectors compute the **cross product**
- If polygon is **convex** all **z components** for the cross products will have the **same sign** (> 0)
- If polygon is **concave** some **z components** for the cross products will have **different sign**



$$(\mathbf{e}_0 \times \mathbf{e}_1)_z > 0$$

$$(\mathbf{e}_1 \times \mathbf{e}_2)_z < 0$$

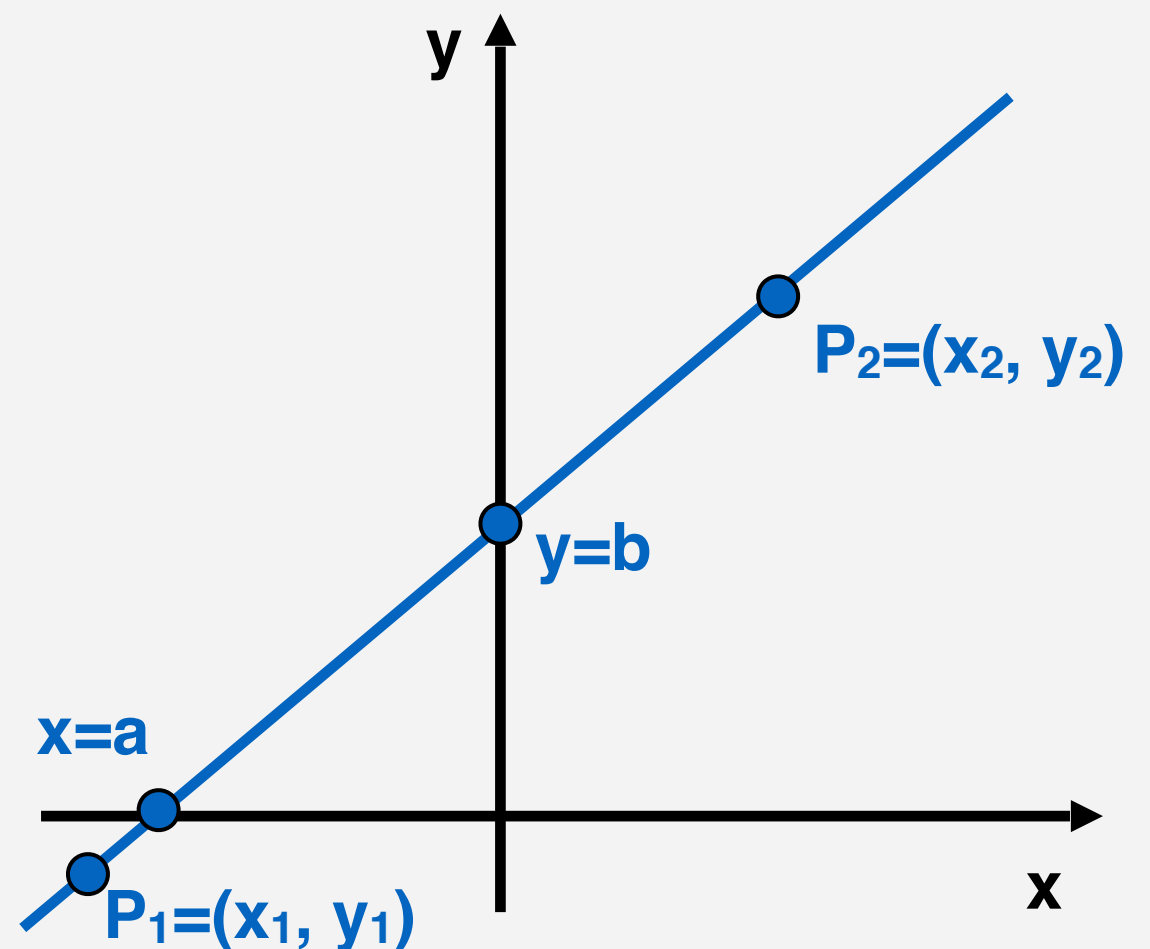
$$(\mathbf{e}_2 \times \mathbf{e}_3)_z > 0$$

Line

“Slope-intercept” equation

- $y = mx + b$
- where m is the slope and the point $(0,b)$ is the *y-intercept*
- cannot be use to express vertical lines

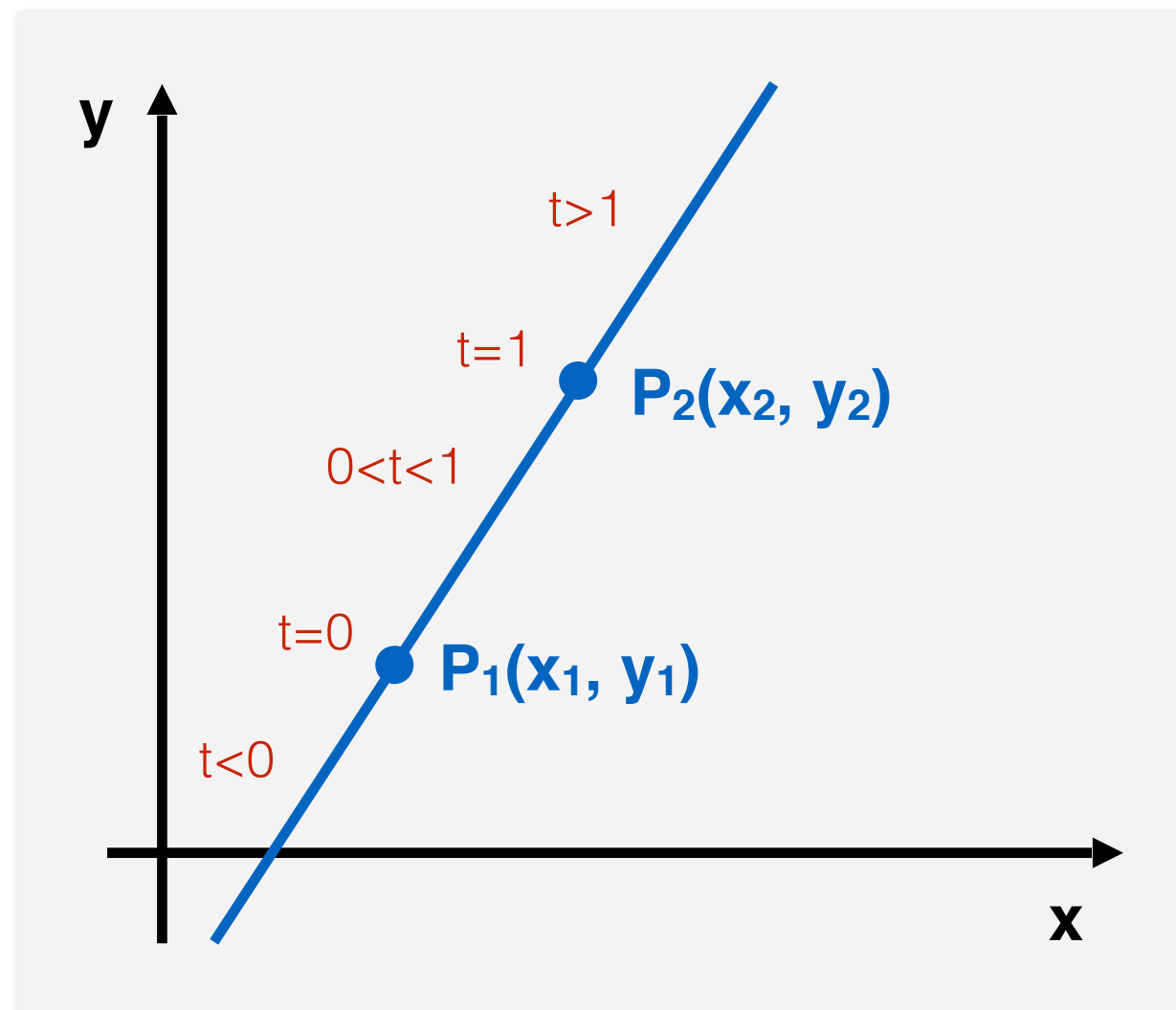
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



Parametric line

$\mathbf{P} = \mathbf{P}_1 + t(\mathbf{P}_2 - \mathbf{P}_1)$ vector based definition

$\mathbf{P} = (1 - t)\mathbf{P}_1 + t\mathbf{P}_2$ affine combination of two points



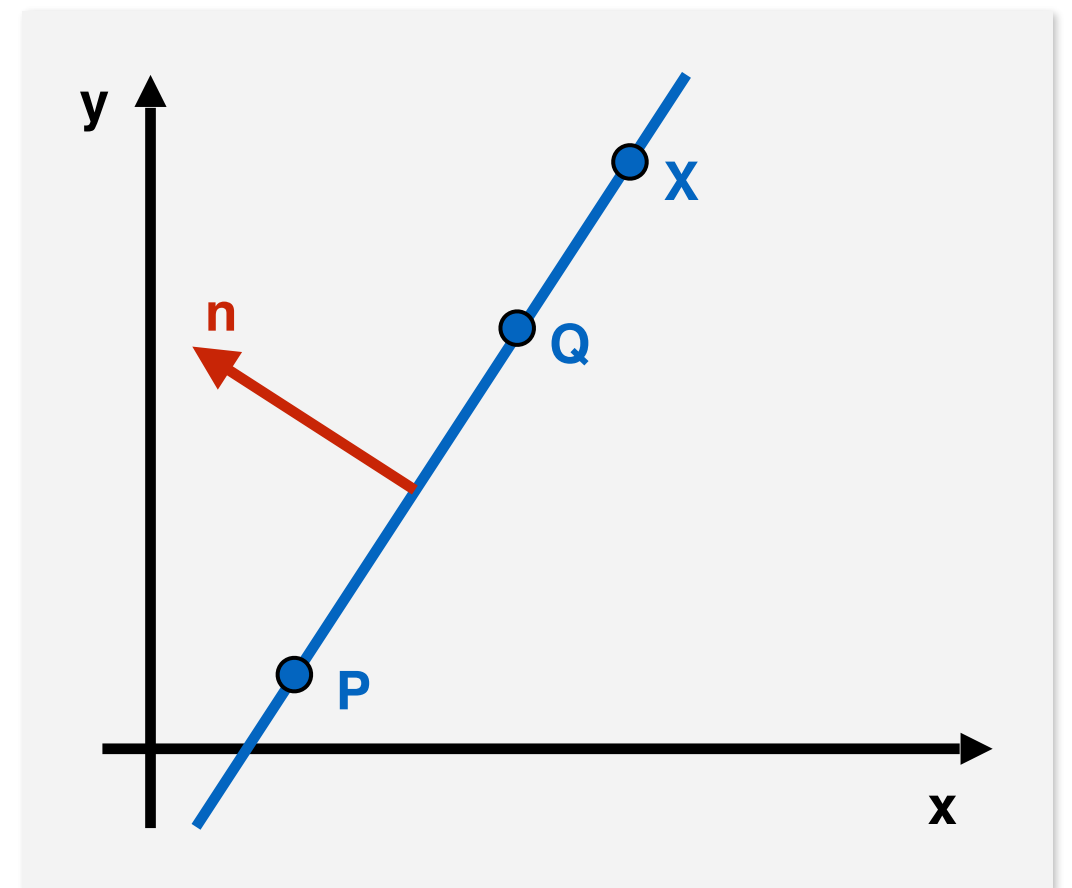
Implicit line

- Vector **n** is perpendicular to the line (called **a normal vector**)
- If X is a point of the line, then

$$(X - P) \cdot \mathbf{n} = 0$$

- The standard implicit form for a line

$$F(X) = (X - P) \cdot \mathbf{n}$$

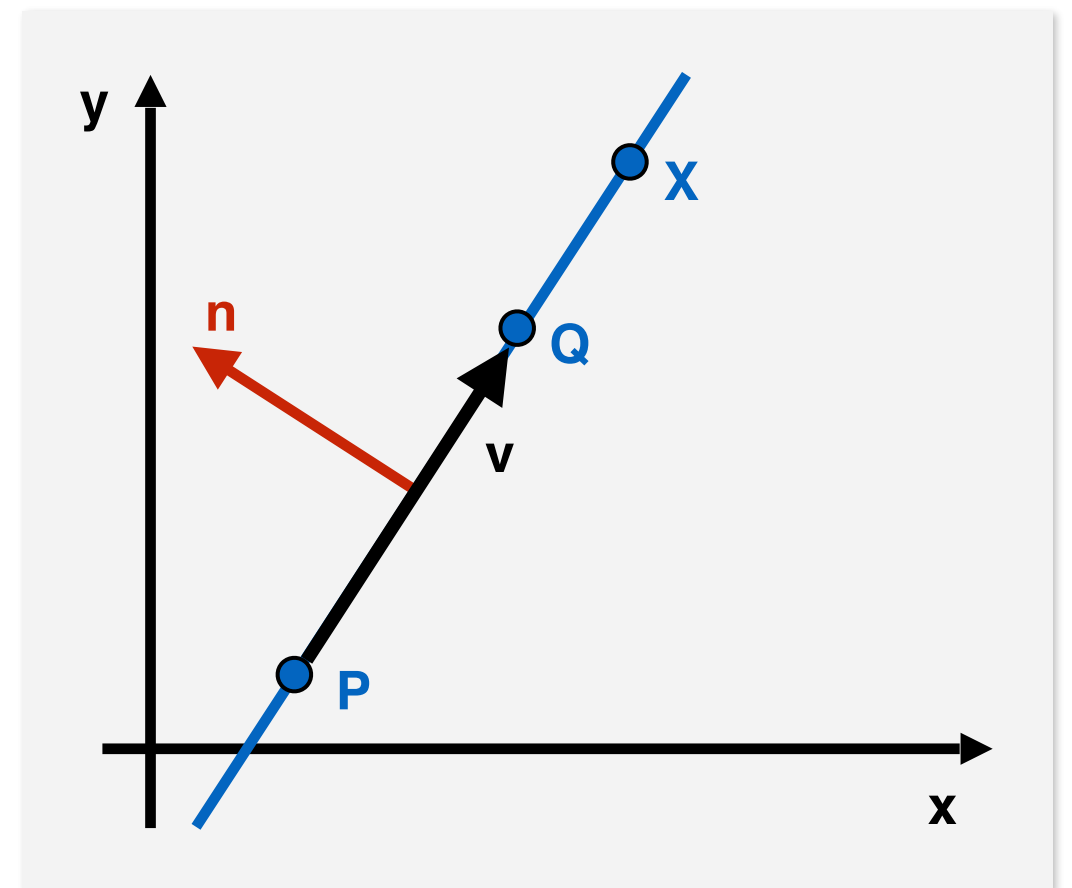


Implicit line

Computing a normal vector to a line:

$$\mathbf{v} = Q - P = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

$$\mathbf{n} = \begin{bmatrix} -v_y \\ v_x \end{bmatrix}$$



Parametric-parametric line intersection

$$\mathbf{P} = (1 - t)\mathbf{P}_1 + t\mathbf{P}_2$$

$$\mathbf{P} = (1 - s)\mathbf{Q}_1 + s\mathbf{Q}_2$$

The equation should have real solution on t and s :

$$(1 - t)\mathbf{P}_1 + t\mathbf{P}_2 = (1 - s)\mathbf{Q}_1 + s\mathbf{Q}_2$$

In fact it is:

$$(1 - t)\mathbf{P}_{x_1} + t\mathbf{P}_{x_2} = (1 - s)\mathbf{Q}_{x_1} + s\mathbf{Q}_{x_2}$$

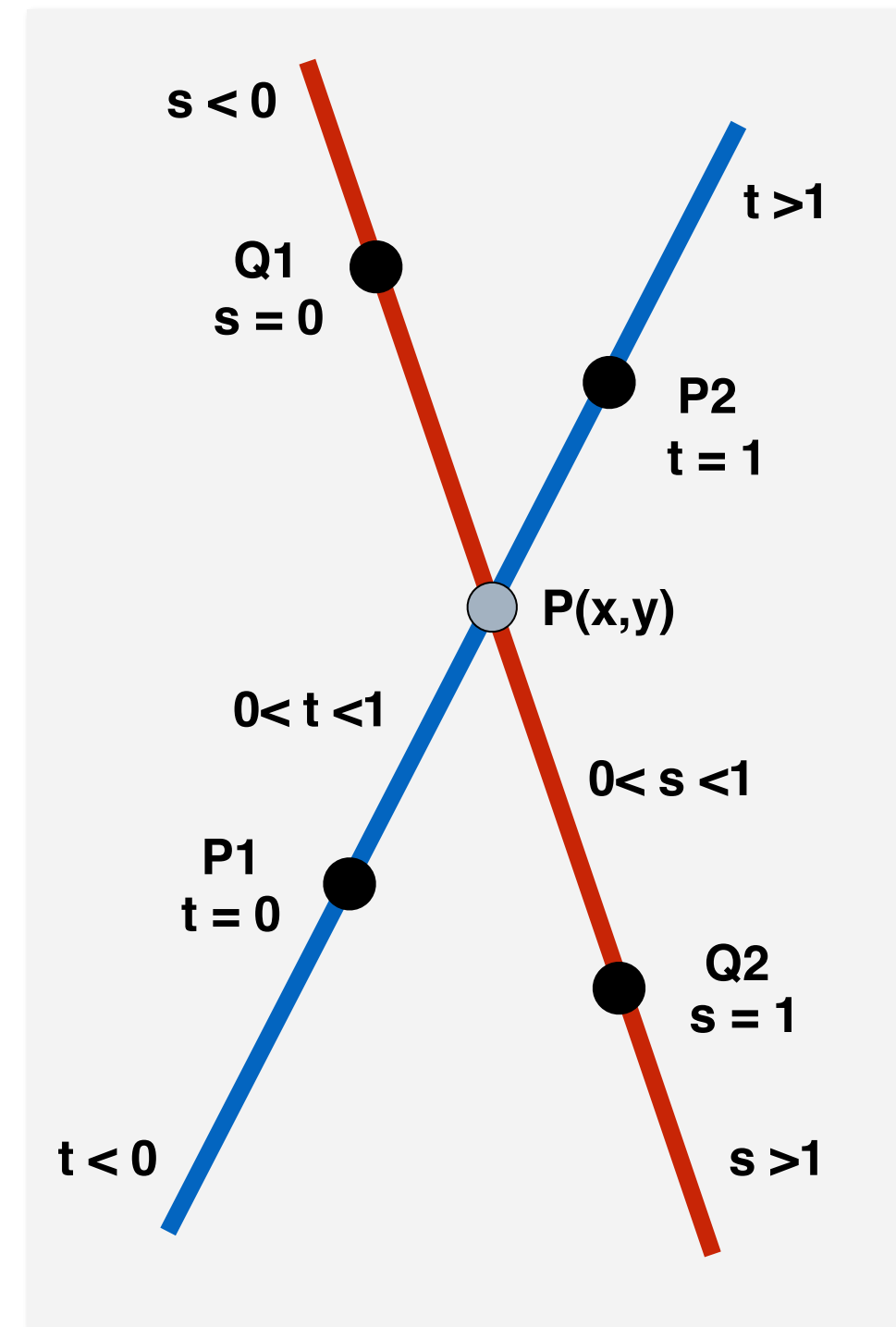
$$(1 - t)\mathbf{P}_{y_1} + t\mathbf{P}_{y_2} = (1 - s)\mathbf{Q}_{y_1} + s\mathbf{Q}_{y_2}$$

Discussion on the $\mathbf{t}$ and $\mathbf{s}$:

$t < 0$, $s < 0$, P before P_1 and Q_1

$0 < t < 1$ and $0 < s < 1$, P_1P_2 intersects Q_1Q_2

$t > 1$, $s > 1$, P after P_2 and Q_2



Parametric-implicit line intersection

$$X = (1 - t)P + tQ$$

$$(X - S) \cdot \mathbf{n} = 0$$

$$((1 - t)P + tQ - S) \cdot \mathbf{n} = 0$$

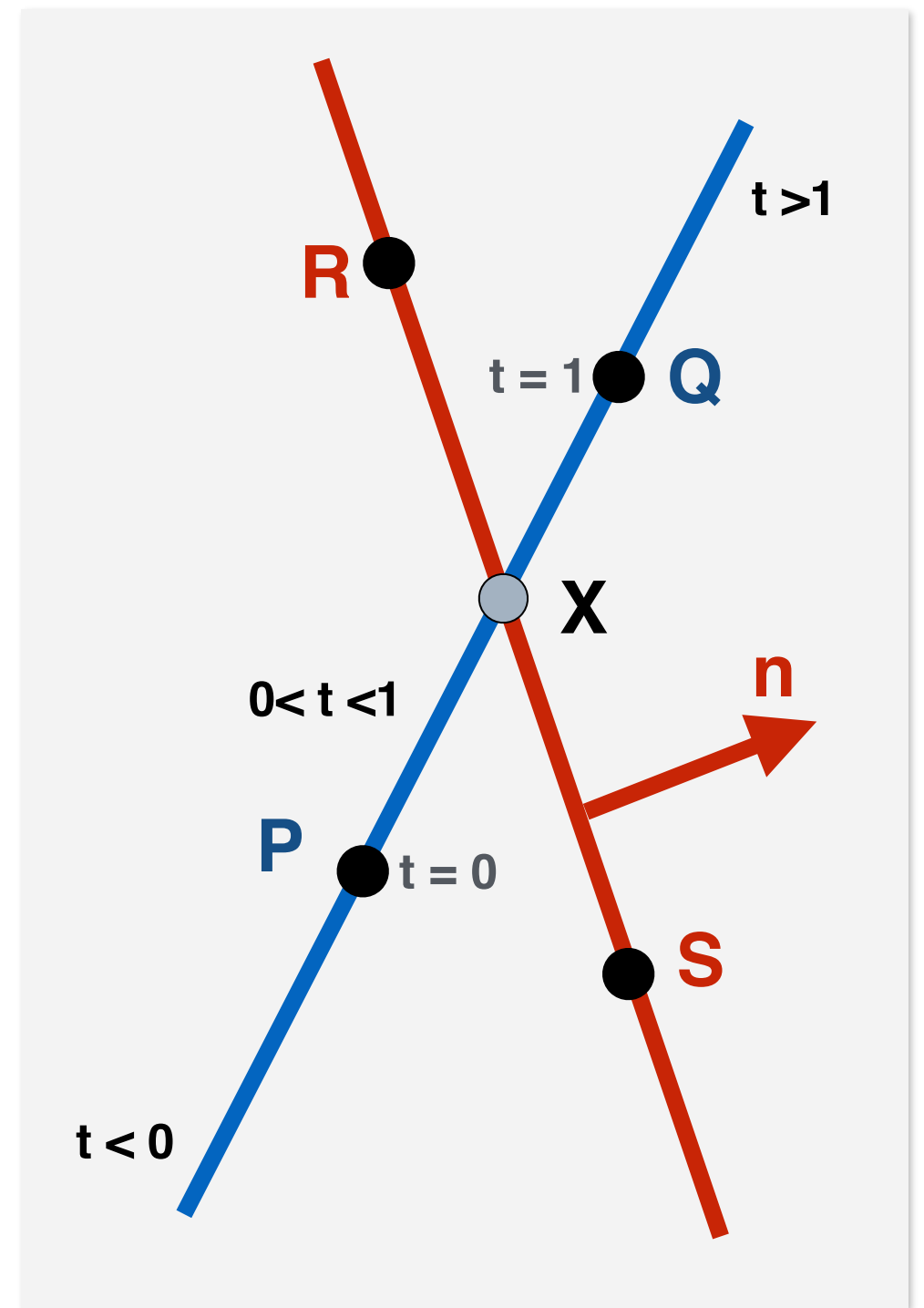
$$(P + t(Q - P) - S) \cdot \mathbf{n} = 0$$

$$((P - S) + t(Q - P)) \cdot \mathbf{n} = 0$$

$$\mathbf{u} = Q - P \quad \mathbf{v} = P - S$$

$$t\mathbf{u} \cdot \mathbf{n} + \mathbf{v} \cdot \mathbf{n} = 0$$

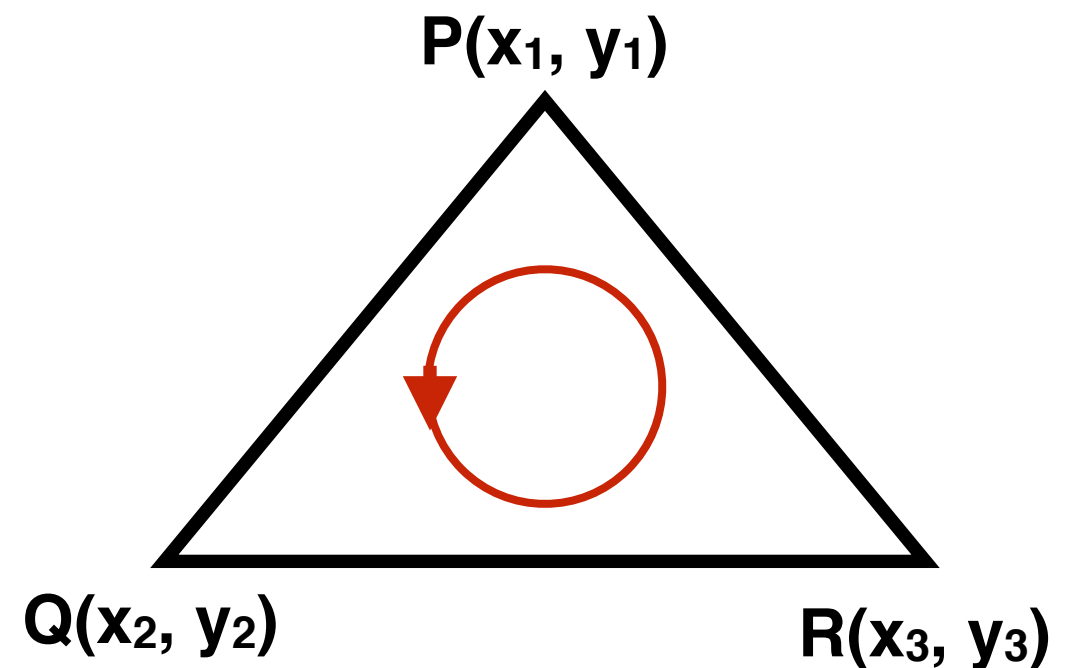
$$t = \frac{-\mathbf{v} \cdot \mathbf{n}}{\mathbf{u} \cdot \mathbf{n}}$$



Point to point relationship

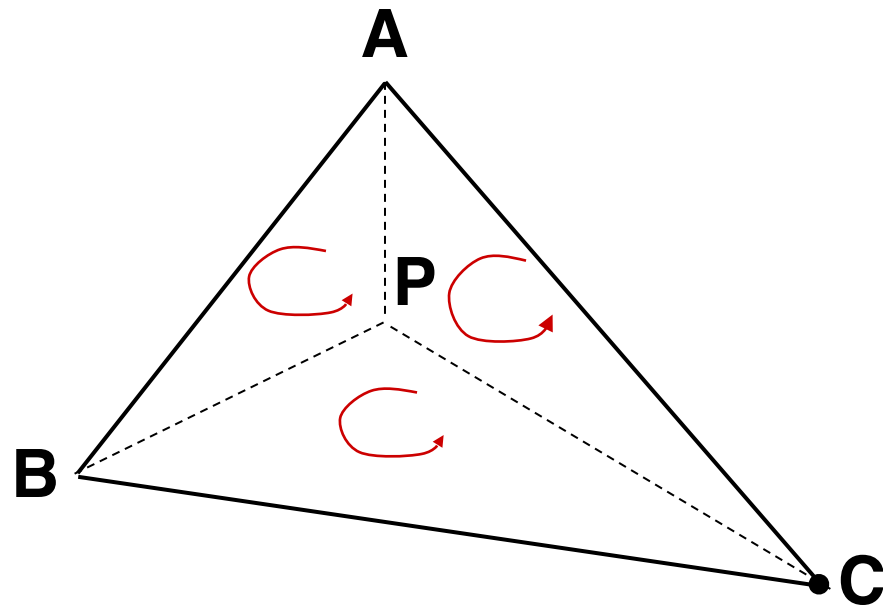
- The order in which three points P, Q, and R are expressed can be determined with the sign of the determinant:

$$D = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

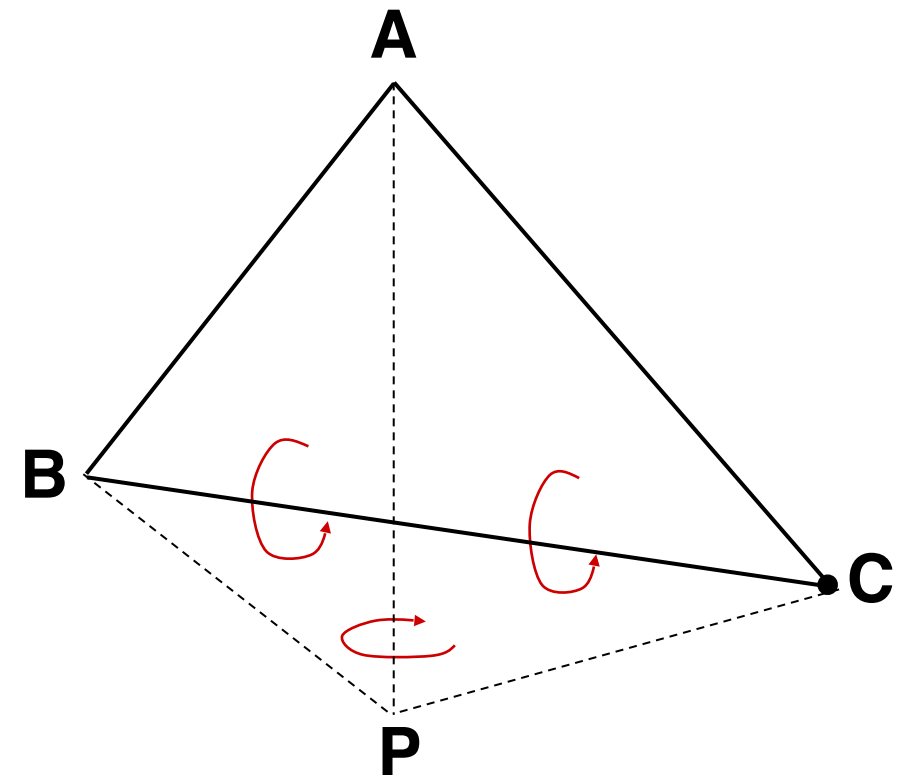


- If **D > 0** then **counterclockwise** direction
- If **D = 0** then **collinear points**
- If **D < 0** then **clockwise** direction

Point to triangle relationship



- **Inside** of the triangle
 - $\text{sign}(A,B,P) = \text{sign}(B,C,P) = \text{sign}(C,A,P)$

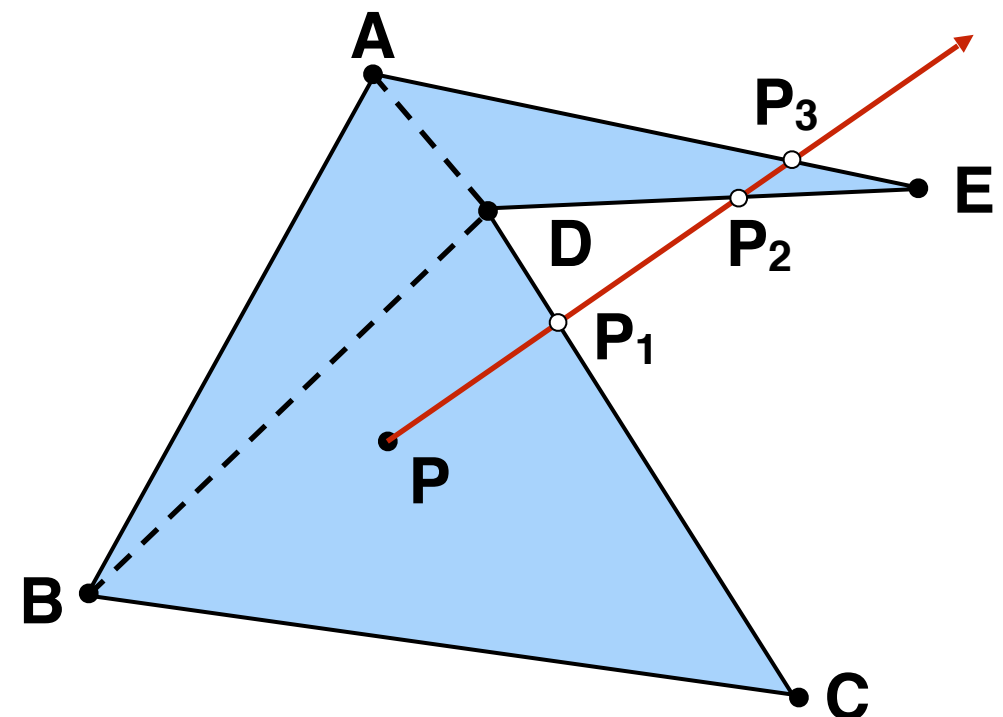
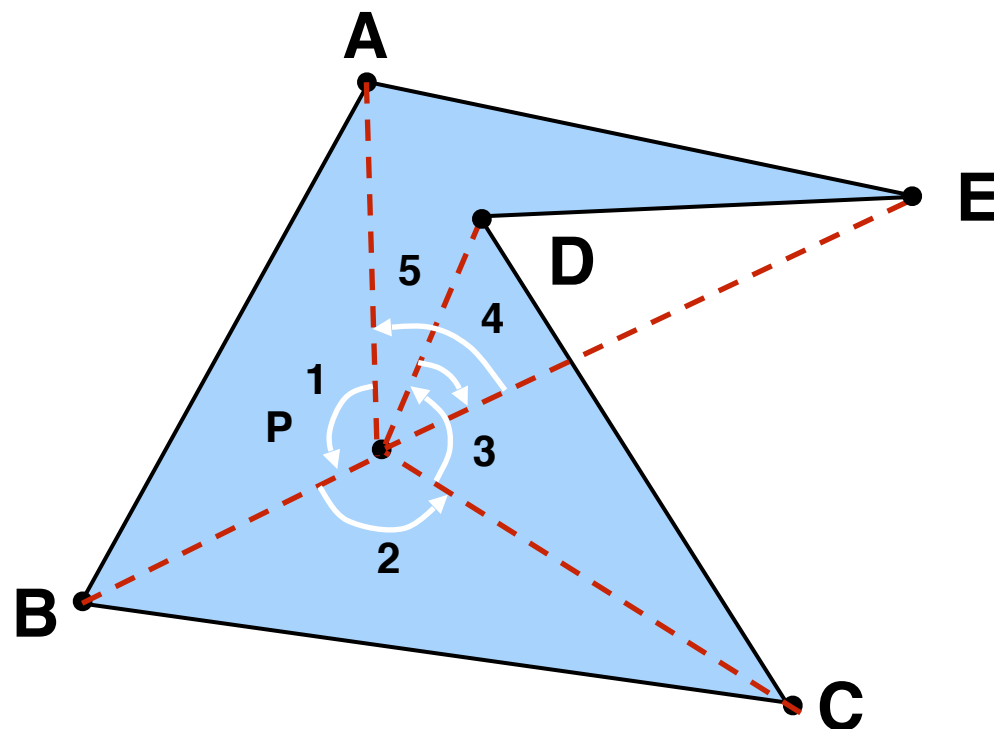


- **Outside** of the triangle
 - NO:
 $\text{sign}(A,B,P) = \text{sign}(B,C,P) = \text{sign}(C,A,P)$

2D Point to polygon relationship

P lays inside the polygon

- Sum of angles given by the edges is 360°
 - $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 = 360^\circ$
- P lays inside at least one triangle of the triangulation set
 - $\triangle ABD, \triangle BDC, \triangle DEA, P \in \triangle ABD$
- Number of intersection between a line starting on P and the triangle
 - Odd number of intersection: P1, P2, P3



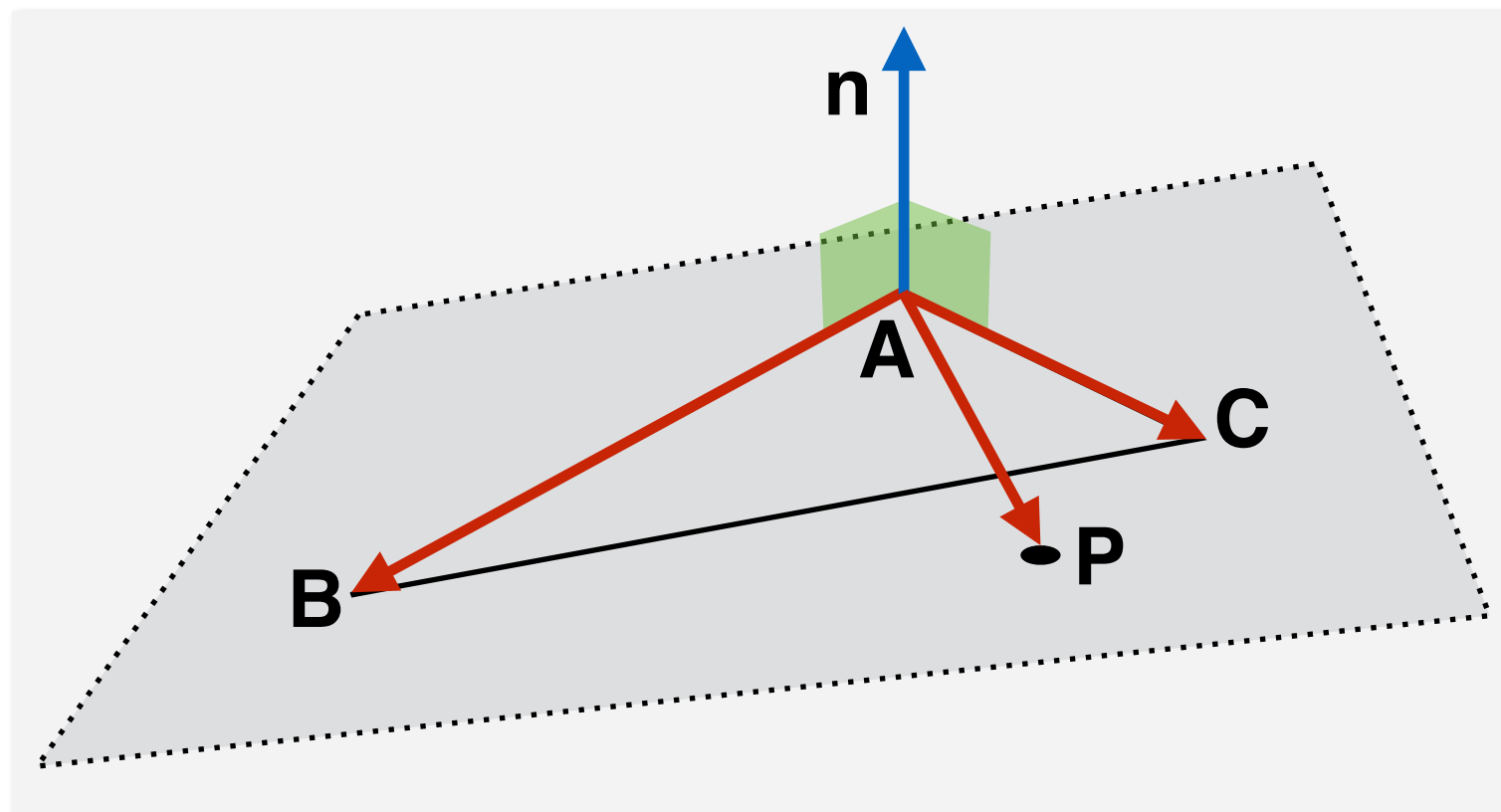
Implicit 3D plane equation

Plane defined by **A**, **B**, and **C** points

The normal is: $\mathbf{n} = (B - A) \times (C - A)$

For every point in the plane: $\mathbf{n} \cdot (P - A) = 0$

Implicit plane equation: $f(P) = \mathbf{n} \cdot (P - A) = 0$

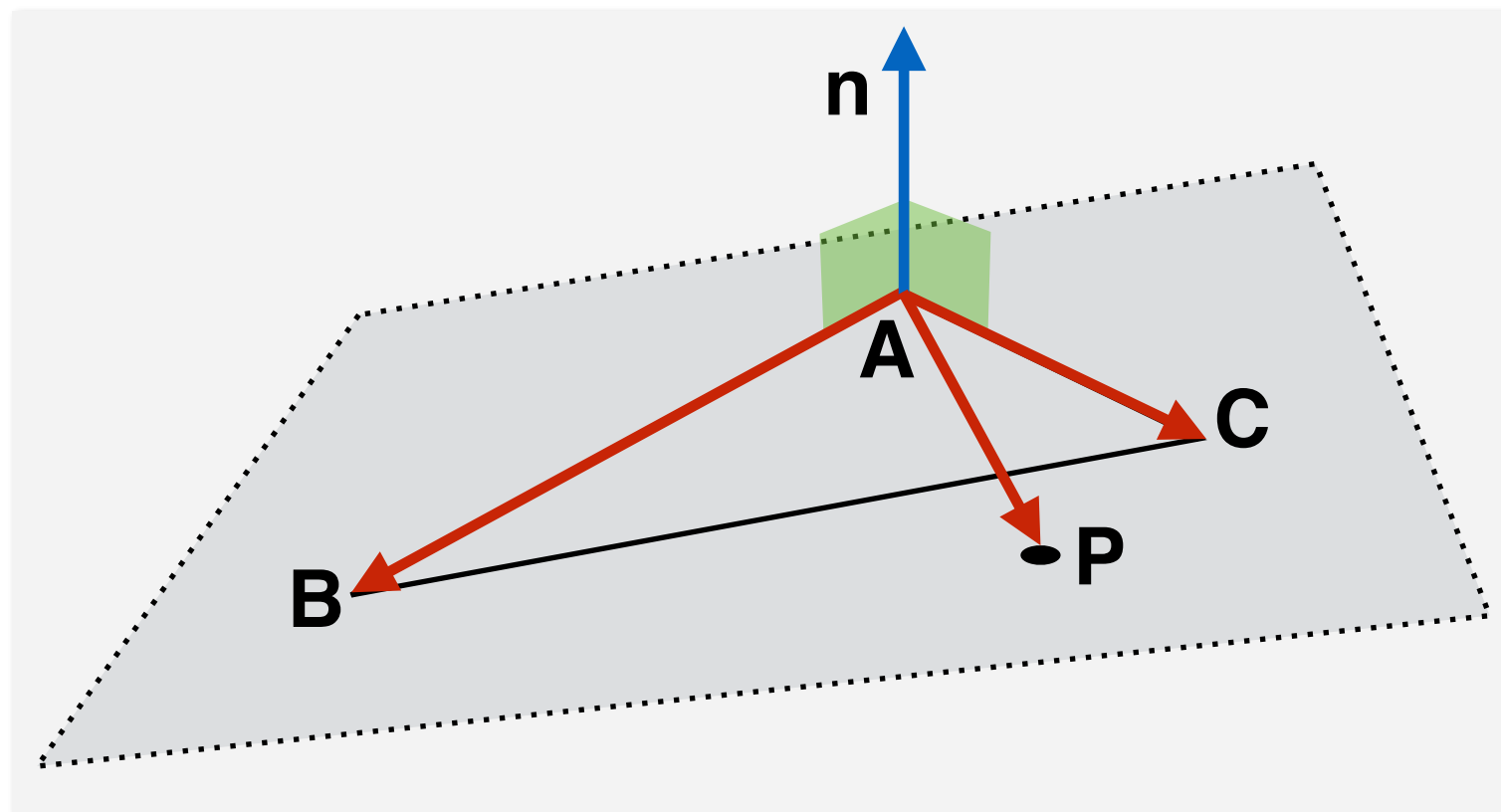


Point to plane relationship

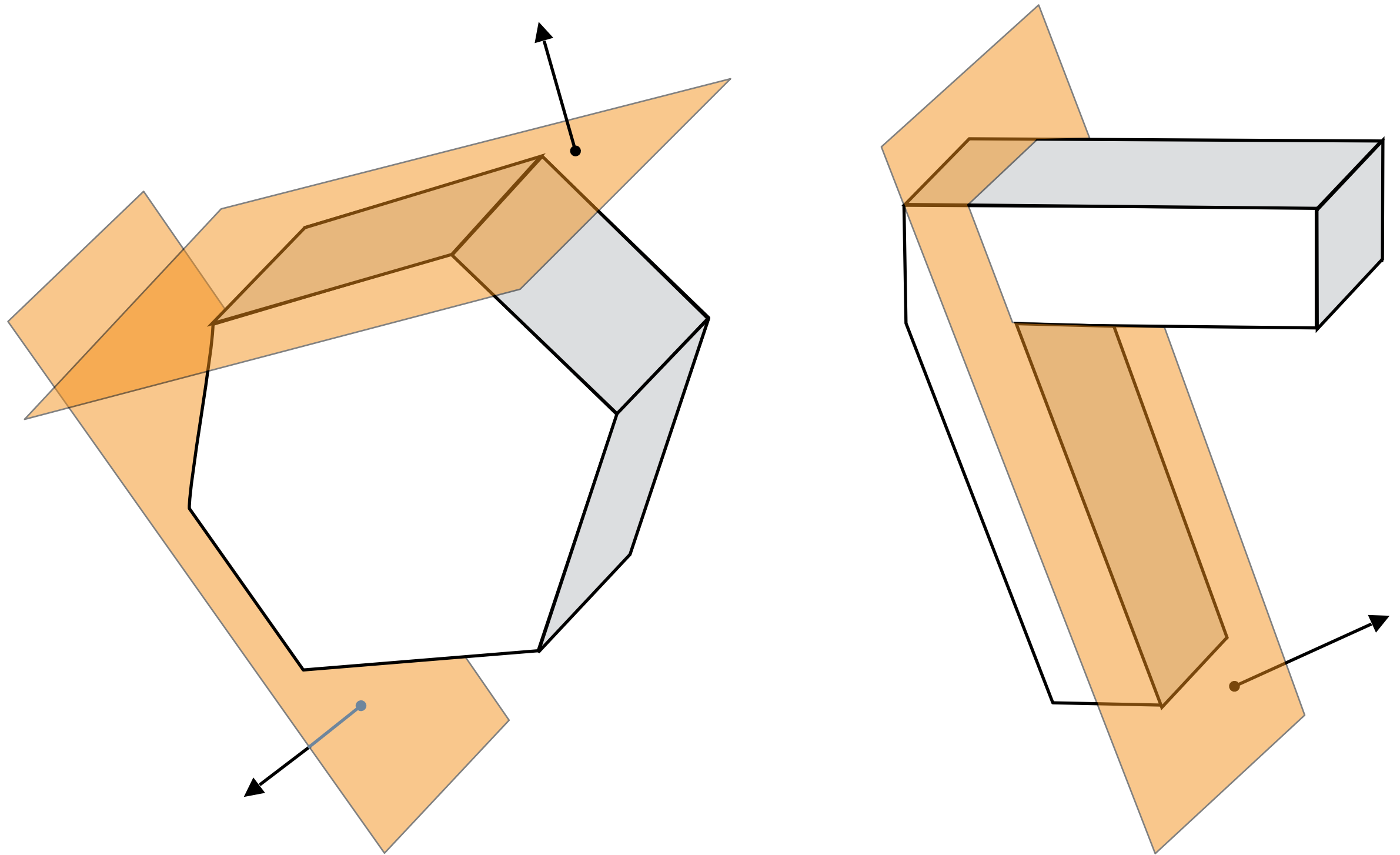
If $f(P) = 0$ then point P is on the plane

If $f(P) > 0$ then point P is on the same side as $\mathbf{n}$

If $f(P) < 0$ then point P is on the other side of $\mathbf{n}$



Convex and concave 3D object



Line - plane intersection

Line equation: $L(t) = P + (Q - P)t$

Plane equation: $(I - A) \cdot \mathbf{n} = 0$

$$(P + (Q - P)t - A) \cdot \mathbf{n} = 0$$

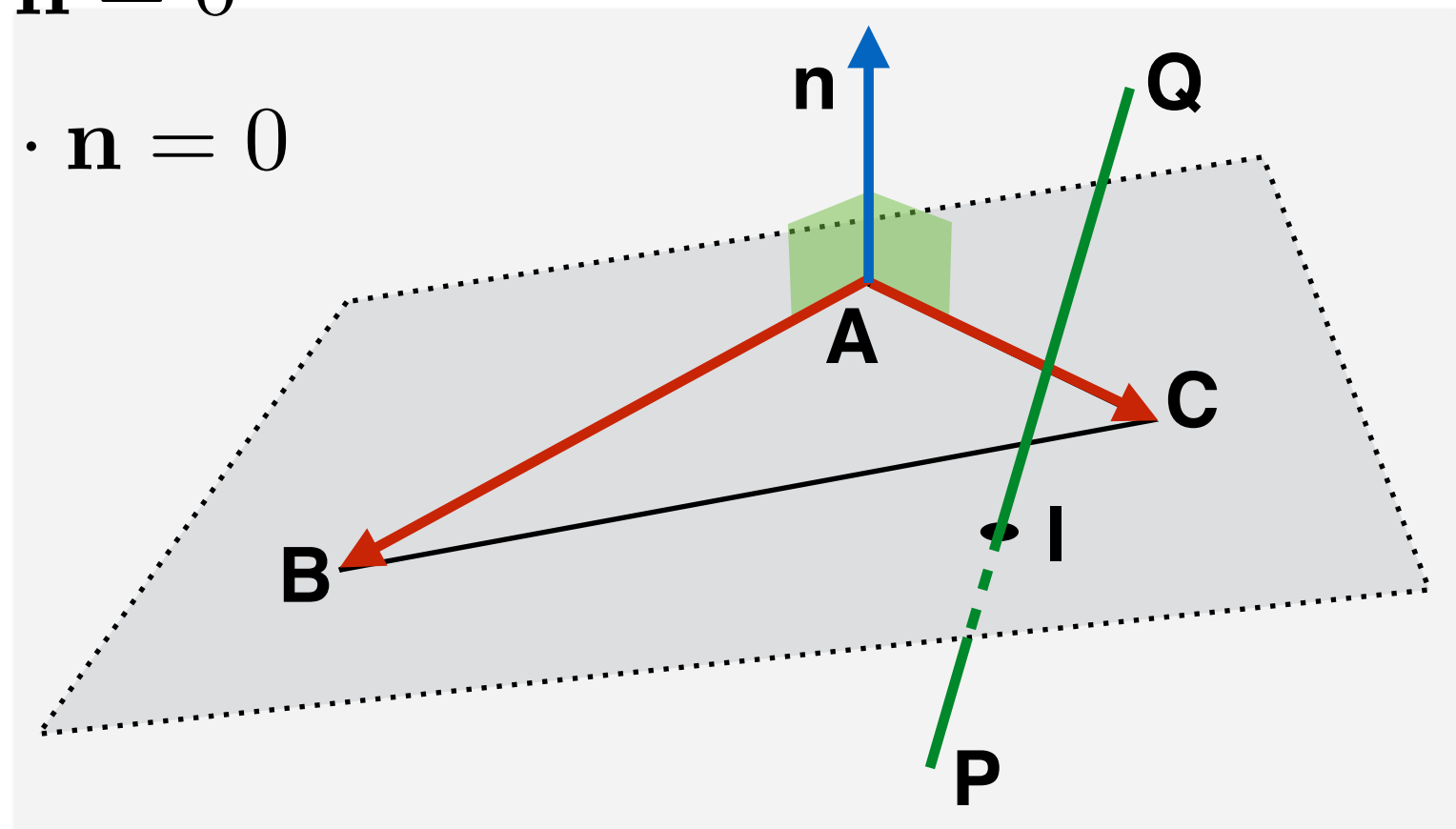
$$((P - A) + (Q - P)t) \cdot \mathbf{n} = 0$$

$$(P - A) \cdot \mathbf{n} + t(Q - P) \cdot \mathbf{n} = 0$$

$$t = \frac{(A - P) \cdot \mathbf{n}}{(Q - P) \cdot \mathbf{n}}$$

Intersection point:

$$I = L(t)$$



Triangle - plane intersection

- Check if a triangle vertex is on the opposite side to the others
- Find the intersection points where the plane cuts the triangle
- Generate 3 new triangles

